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Multiobjective Optimization of Suction Cup and Parallel Jaw Grippers

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Abstract

This master's thesis investigates multi-objective optimization methods for the design of robotic grippers used in bin-picking applications. The aim is to compare selected optimization algorithms on two gripper-design problems: the placement of three suction cups and the design of rigid parallel-jaw contours. Both problems are treated as black-box optimization tasks with multiple, potentially conflicting objectives.

First, an overview of robotic grasping, gripper design, and multi-objective optimization is presented. The principles of evolutionary algorithms, Bayesian optimization, and performance indicators are then introduced. A literature review examines existing approaches to grasp planning, gripper design, and algorithm comparison.

Subsequently, the two gripper-design problems are formulated and integrated into a common optimization framework. Suction-cup layouts are assessed using grasp diversity, the number of retained grasps, and the area spanned by the cups. Parallel-jaw contours are represented by splines and evaluated using geometric proximity at four predefined workpiece orientations.

NSGA-II, NSGA-III, MOEA/D, SMS-EMOA, and CMOPSO are compared on four workpieces. Their results are assessed using inverted generational distance plus, hypervolume, and runtime. For the suction-cup problem, SMS-EMOA achieves the lowest mean inverted generational distance plus on all four workpieces. For the parallel-jaw problem, NSGA-II achieves the highest mean archive hypervolume on all four workpieces.

Bayesian optimization is also investigated. The main attempt is stopped because of its runtime and produces no final result files. A subsequent single-run test on the parallel-jaw gripper and connecting rod provides indicative results but does not support a conclusive comparison.

Finally, the limitations of the study and directions for future work are discussed. These include further Bayesian experiments, improved reproducibility, non-overlap constraints for the suction cups, and physical validation of the optimized gripper designs.

Keywords: multi-objective optimization, robotic gripper design, bin picking, suction-cup gripper, parallel-jaw gripper, evolutionary algorithms, Bayesian optimization

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List of Abbreviations

3D	Three-dimensional
AI	Artificial Intelligence
API	Application Programming Interface
BBOB	Black-Box Optimization Benchmarking
BO	Bayesian Optimization
CMOPSO	Competitive Multi-Objective Particle Swarm Optimizer
CPU	Central Processing Unit
DFT	Discrete Fourier Transform
DTLZ	Deb–Thiele–Laumanns–Zitzler test problems
EA	Evolutionary Algorithm
EI	Expected Improvement
FFT	Fast Fourier Transform
FPS	Farthest-Point Sampling
GP	Gaussian Process
GPU	Graphics Processing Unit
HV	Hypervolume
IGD	Inverted Generational Distance
IGD+	Inverted Generational Distance Plus
IPA	Fraunhofer Institute for Manufacturing Engineering and Automation
JSON	JavaScript Object Notation

KPI	Key Performance Indicator
MBM	Modular BoTorch Model
MCB	Mechanical Components Benchmark
MOEA	Multi-Objective Evolutionary Algorithm
MOEA/D	Multi-Objective Evolutionary Algorithm based on Decomposition
MOO	Multi-Objective Optimization
MOP	Multi-Objective Optimization Problem
NFL	No Free Lunch
NSGA-II	Non-dominated Sorting Genetic Algorithm II
NSGA-III	Non-dominated Sorting Genetic Algorithm III
PC	Personal Computer
PF	Pareto Front
PJ	Parallel-Jaw
PSO	Particle Swarm Optimization
qLogNEHVI	q-Log Noisy Expected Hypervolume Improvement
qNEHVI	q-Noisy Expected Hypervolume Improvement
RAM	Random-Access Memory
RBF	Radial Basis Function
RGB-D	Red, Green, Blue and Depth
SC	Suction Cup
SD	Standard Deviation
SMS-EMOA	S-Metric Selection Evolutionary Multi-Objective Algorithm
TCP	Tool Center Point
UML	Unified Modeling Language
ZDT	Zitzler–Deb–Thiele test problems

Nomenclature

α_i Angular position of suction cup i, $i = 1,2,3$

$\alpha_{\mathrm{EI}}(\mathbf{x})$ Expected improvement acquisition

Δp Pressure difference

ℓ_{gn} Length of the spline-based gripping contour

γ_i Grasp i of the predefined grasp set

$\hat{A}(\cdot)$ Normalized triangle-area objective of a gripper layout

$\hat{n}_{\mathrm{g}}(\cdot)$ Normalized grasp count of a gripper layout, i.e. the number of evaluated grasps divided by the clustering budget

$\kappa(\mathbf{x}, \mathbf{x}')$ Covariance/kernel function

$\mathbf{f}(\mathbf{x})$ Objective function vector

$\mathbf{q}_{\mathrm{PJ}}$ Four-dimensional decision vector defining the parallel-jaw gripper contour

$\mathbf{q}_{\mathrm{SC}}$ Four-dimensional decision vector defining the suction-cup gripper configuration

$\mathbf{x}$ Decision variable vector

$\mathbf{x}^{(a)} \prec \mathbf{x}^{(b)}$ Dominance relation (solution a dominates b)

$\mathbf{z}_{\mathrm{ref}}$ Hypervolume reference point

$\mathcal{D}$ Observed data set

$\mathcal{GP}$ Gaussian process

$\mathcal{R}$ Set of predefined workpiece orientations of a two-finger optimization run

$\mathcal{W}_i$ Workpiece geometry in the orientation of grasp γ_i

$\mathcal{X}$ Feasible decision space

$\mathrm{HV}(A, \mathbf{z}_{\mathrm{ref}})$ Hypervolume indicator

$\mu(\mathbf{x})$ Predictive mean

$\sigma^2(\mathbf{x})$ Predictive variance

$A(\cdot)$ Area of the triangle formed by the three suction cups of a gripper layout

A_{seal} Effective sealed area

$A_{\max}$ Global normalization constant: area of the equilateral triangle inscribed in a circle with $r_{\max}$ set to the Stanford Bunny value (0.048 m), the largest among the four workpieces

d_{geo} Geodesic distance on $SO(3)$

D_{pos} Normalized spatial diversity

D_{rot} Normalized orientational diversity

$f_i(\mathbf{x})$ Individual objective function

F_{hold} Holding force

$f_{\min}$ Best observed objective value

h_j Spline depth parameter at relative contour position $j/4$, $j = 1,2,3$

$k_i(\cdot)$ Grasp-quality KPI (`max_prox`) of grasp γ_i

n_{or} Number of predefined grasp orientations of a two-finger optimization run

p_{atm} Atmospheric pressure

r Common radial distance of the suction-cup centers from the gripper origin

r_{cup} Suction-cup radius

r_{max} Upper radial bound for suction-cup placement

r_{min} Lower radial bound for suction-cup placement

S Safety factor

$SO(3)$ Special orthogonal group of 3D rotations

1 Introduction

1.1 Motivation

Robotic grasping is a fundamental capability in automated manufacturing, logistics, and service robotics [6]. The ability to reliably grasp objects of varying shapes, sizes, and surface properties remains a challenging problem, particularly in unstructured environments [7]. Industrial applications such as bin picking, assembly, and material handling require grippers that can operate robustly across a wide range of workpieces [8].

Among the commonly used end-effectors, suction-cup grippers and two-finger grippers represent two complementary approaches [9]. Suction grippers are widely used for handling objects with smooth surfaces due to their simplicity and speed, whereas two-finger grippers provide flexibility in handling objects with complex geometries [9, 10]. Designing optimal configurations for these grippers is non-trivial, as grasp performance depends on multiple interacting factors such as geometry, contact distribution, force limits, and collision constraints [11].

While the gripper configuration problems are well-motivated in themselves, the primary objective of this thesis is not simply to find the best gripper configuration, but to systematically apply and compare optimization algorithms for solving these configuration problems. A large number of optimization algorithms exist, spanning evolutionary strategies, decomposition-based methods, indicator-based selection, and Bayesian optimization, yet their relative suitability for gripper design problems of this type is not well understood [12]. Different algorithms may exhibit substantially different convergence behavior, solution quality, and robustness depending on the structure of the objective landscape and the dimensionality of the search space. Identifying which algorithm class performs best for each gripper problem is therefore a practically relevant and scientifically meaningful research question. [12, 13]

1.2 Problem Definition

The design of task-specific robotic grippers remains largely a manual and experience-driven process. Geometric parameters, such as the placement of suction cups or the contour of gripper jaws, are commonly selected iteratively and subsequently validated through simulation, geometric evaluation, or physical experiments [14]. This procedure can become time-consuming when many possible configurations must be considered.

For the gripper-design problems investigated in this thesis, the performance of a candidate configuration is determined by computational evaluation routines. These routines do not provide analytical gradients or closed-form expressions that directly relate the design parameters to the resulting performance. Furthermore, a gripper design must generally satisfy several criteria simultaneously, for example by performing across multiple workpieces, grasp locations, or geometric requirements. Improvements with respect to one criterion may reduce performance with respect to another. Representing all criteria by a single aggregated value could hide these trade-offs.

These characteristics lead to expensive, derivative-free, multi-objective black-box optimization problems [15]. A variety of optimization approaches can be applied to such problems, including evolutionary multi-objective algorithms and Bayesian optimization methods. However, their performance depends on factors such as the problem representation, number of objectives, objective landscape, and available evaluation budget [16, 17]. Consequently, results obtained for generic test problems or other applications cannot necessarily be transferred directly to task-specific gripper design.

The primary goal of this thesis is therefore to apply and comparatively evaluate selected multi-objective evolutionary algorithms for automated gripper configuration, and to examine the practical applicability of multi-objective Bayesian optimization under the computational conditions of the investigated problems. Two application cases are considered: the configuration of a multi-suction-cup gripper and the optimization of rigid jaw contours for a two-finger gripper. In the first case, the positions of the suction cups are optimized to obtain layouts that perform well for a given workpiece. In the second case, spline-based jaw contours are optimized to provide suitable geometric conformity at predefined grasp points. The two cases provide different geometric representations and objective structures through which the behavior of the selected optimization algorithms can be investigated.

The objective is not to identify a universally superior optimization algorithm. Instead, the thesis examines how the selected methods perform on the two specific gripper-design

problems under comparable experimental conditions. The comparison considers the quality of the obtained multi-objective solutions, their development over the available evaluation budget, and the consistency of the results across repeated runs.

To achieve this goal, the following tasks are defined:

- **Problem formulation:** Formulate the multi-suction and two-finger gripper configuration problems as multi-objective black-box optimization tasks by specifying their decision variables, objectives, and applicable constraints.
- **Framework integration:** Establish a common optimization interface through which the selected algorithms can access both gripper-evaluation pipelines under consistent conditions.
- **Evaluation preparation:** Integrate and adapt the problem-specific evaluation procedures required to assess candidate suction-cup layouts and rigid jaw contours.
- **Experimental design:** Define representative workpieces, predefined evaluation configurations, evaluation budgets, initialization conditions, repeated runs, and performance indicators for a controlled and reproducible comparison.
- **Algorithm application and comparison:** Apply selected evolutionary and Bayesian multi-objective optimization methods to both gripper-design problems and compare their performance using suitable multi-objective indicators.
- **Result analysis:** Analyze the obtained results with respect to solution quality, convergence behavior, robustness across repeated runs, and sensitivity to the characteristics of the investigated gripper-design problems.

2 Theoretical Background

2.1 Robotics and Object Handling

2.1.1 Robotics

An industrial robot is an automatically controlled, reprogrammable multipurpose manipulator(s), programmable in three or more axes, which can be either fixed in place or fixed to a mobile platform for use in automation applications in an industrial environment [18]. A typical robotic system consists of a kinematic chain of rigid links connected by actuated joints, a set of sensors, a controller and an end-effector mounted at the flange of the last link. The configuration of the manipulator is described by its joint variables, while the position and orientation of the end-effector are described in a Cartesian task space [19]. The mapping between these two spaces is established through forward and inverse kinematics, which form the basis for trajectory planning and motion control [20].

Modern manufacturing systems increasingly require flexibility to accommodate changing production demands, mass customization, increasing product variants, and decreasing batch sizes [21, 22]. Robot-cell reconfiguration and programming nevertheless remain time-consuming and often require specialist expertise [21]. Task-specific assets such as grippers, sensors, and actuators provide robot cells with the capabilities required for particular manufacturing tasks [21].

2.1.2 Bin Picking Problem

Bin picking describes the task of removing individual workpieces from an unordered accumulation of parts, typically supplied in a container, and transferring them to a defined target pose [23, 24]. The task combines perception, grasp planning, collision-free motion planning and physical grasp execution. Because the parts are randomly oriented and may be partially

occluded or entangled, the pose of the next object to be grasped is not known in advance and must be estimated from sensor data at runtime [25, 26].

For gripper design, the decisive consequence of this setting is that a grasp cannot be planned for one privileged object pose [23]. The gripper must be able to reach and hold the workpiece across a broad range of orientations and partial accessibility conditions, which implies that a useful gripper configuration provides several alternative grasp candidates distributed over the object surface rather than one single optimal contact [23, 27]. Additional constraints arise from the confined working space inside the bin, where collisions between the gripper body and the bin walls or neighboring parts restrict the set of feasible approach directions [26, 28]. These requirements directly motivate the objectives used in this thesis, in particular the diversity of the available grasps for the suction cup gripper and the geometric conformance of the finger contour for the two-finger gripper.

2.2 Gripper Theory and Geometric Representation

2.2.1 Grasp Theory

A grasp is a configuration of contact points between a gripper and a workpiece such that the gripper can resist external forces and torques applied to the object [29]. The classical criterion for grasp quality is *force closure*. A grasp is force-closed if the contact forces available at the contact points can produce any net wrench on the object, meaning that arbitrary disturbances can be resisted [30]. Force closure requires a sufficient number of contacts with appropriate friction properties and geometric placement.

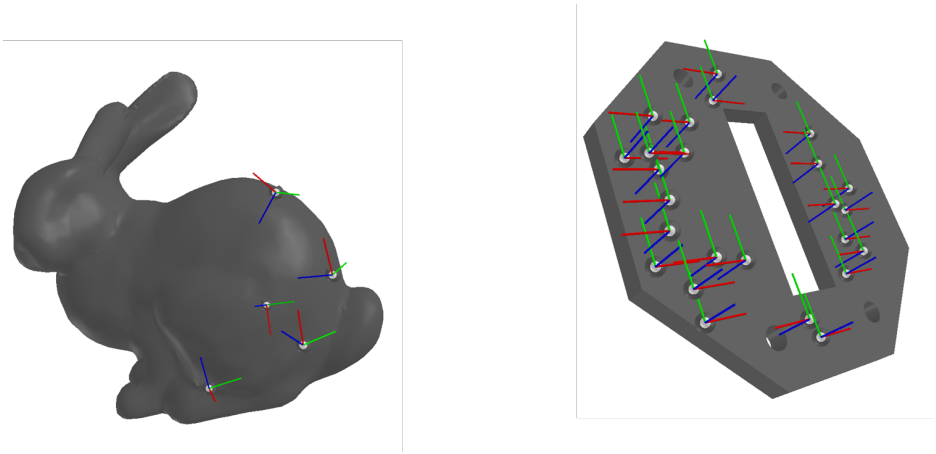


Figure 2.1: Visualization of a grasp configuration on workpieces.

Figure 2.1 illustrates grasp configurations for two workpieces: the Stanford Bunny and the IPA Intermediate Plate. Each point on the workpiece surface encodes the translation and rotation required to transform the gripper's Tool Center Point (TCP) from its initial pose to the corresponding grasp pose.

2.2.2 Suction Cup Grippers

Suction cup grippers belong to the astrictive gripper class and generate holding force from a pressure difference between the evacuated volume inside the cup and the ambient atmosphere. The holding force is defined as:

$$F_{\text{hold}} = \frac{\Delta p \cdot A_{\text{seal}}}{S}, \qquad \Delta p \in (0, p_{\text{atm}}] \subset \mathbb{R}_{>0}, \quad A_{\text{seal}} \in \mathbb{R}_{>0}, \quad S \in [1, \infty) \subset \mathbb{R}, \quad F_{\text{hold}} \in \mathbb{R}_{>0} \tag{2.1}$$

where Δp is the pressure difference between the evacuated cup volume and the ambient atmosphere, bounded above by the atmospheric pressure p_{atm} since a perfect vacuum cannot be exceeded, A_{seal} is the effective sealed area enclosed by the cup lip, and $S \geq 1$ is a safety factor accounting for acceleration, orientation and friction conditions [31, 32]. The holding force is strictly positive for any non-vanishing pressure difference acting on a non-degenerate sealed area. The decisive requirement is therefore an airtight seal between the cup lip and the workpiece surface, which restricts applicable surfaces to regions that are sufficiently

smooth, non-porous and locally flat or only moderately curved relative to the compliance of the cup.

Practical implementations frequently use several cups arranged on a common mounting plate. Distributing the load over multiple cups increases the total holding force, provides redundancy if individual cups fail to seal, and, most importantly for handling, generates a resistance to tilting moments [33]. The moment capacity depends on the spatial arrangement of the active cups, which is commonly characterized through the area of the polygon or triangle spanned by the sealing contacts [32]. A large spanned area yields a stable grasp, whereas narrowly clustered cups behave similarly to a single contact. The placement of the cups on the mounting plate is thus the central design variable, and it must be chosen such that valid sealing contacts are achievable on every workpiece of the considered family.[34]

2.2.3 Multi-Finger Grippers

Multi-finger grippers are impactive grippers that transmit forces through direct contact between movable jaws and the workpiece. They are classified by the number of fingers, the kinematics of the jaw motion and the number of actuated degrees of freedom [35]. Two-finger parallel grippers with a single actuator represent the industrial standard because of their mechanical simplicity, robustness and low control effort [36]. Three-finger centric designs provide self-centring behavior for rotationally symmetric parts, while anthropomorphic hands with many degrees of freedom offer high versatility at the cost of complexity [37, 38].

Two grasp modes are commonly distinguished. Force-closure grasps hold the object through friction at the contact surfaces and therefore depend on the applied clamping force and the friction coefficient. Form-closure grasps immobilize the object through the geometry of the contacts alone, so that no disturbance can move the object without deformation [39]. Because a mechanically simple gripper provides few actuated degrees of freedom, the ability to realize form closure is shifted from the kinematics to the geometry of the jaws. This is the reason why application-specific jaw inserts, frequently produced additively, are widely used in industry and why the jaw contour is the natural design variable for optimization. [40]

2.3 Optimization Theory

2.3.1 Black-Box Optimization

A black-box optimization problem is one in which the objective function has no known analytical form and no gradient information is available to the optimizer [41]. The function can only be queried at candidate input points, returning a scalar or vector output. This setting arises naturally in engineering design when objectives are evaluated through computationally expensive procedures, making gradient computation either impossible or impractical. Because each function evaluation may be costly, black-box optimizers must find high-quality solutions within a limited evaluation budget, placing emphasis on sample efficiency. [42, 43]

2.3.2 Multi-Objective Optimization

Multi-objective optimization (MOO) refers to the simultaneous optimization of two or more conflicting objective functions over a shared decision space [44]. Formally, a multi-objective minimization problem is defined as

$$\begin{aligned} &\min_{\mathbf{x}\in\mathcal{X}} \mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_k(\mathbf{x})), \\ &\mathcal{X} \subseteq \mathbb{R}^d, \quad d \in \{n \in \mathbb{Z} \mid n \geq 1\}, \quad k \in \{n \in \mathbb{Z} \mid n \geq 2\}, \\ &f_i : \mathcal{X} \to \mathbb{R}, \quad \forall\, i \in \{1, \ldots, k\}, \qquad \mathbf{f}(\mathbf{x}) \in \mathbb{R}^k, \quad \forall\, \mathbf{x} \in \mathcal{X}. \end{aligned} \tag{2.2}$$

In Eq. 2.2, $\mathbf{x}$ is the decision variable vector, $\mathcal{X}$ is the feasible decision space, and $k \geq 2$ is the number of objectives. When objectives conflict, no single solution minimizes all objectives simultaneously. Instead, a set of trade-off solutions exists, collectively forming the Pareto front. The goal of a multi-objective optimizer is therefore not to find a single best solution but to approximate this trade-off surface as accurately and completely as possible [44, 45].

2.3.3 Pareto Optimality

The concept of Pareto optimality provides the theoretical foundation for comparing solutions in a multi-objective setting. A solution $\mathbf{x}^{(a)}$ is said to *dominate* $\mathbf{x}^{(b)}$, written $\mathbf{x}^{(a)} \prec \mathbf{x}^{(b)}$, if and only if

$$f_i(\mathbf{x}^{(a)}) \leq f_i(\mathbf{x}^{(b)}) \quad \forall\, i \in \{1,\ldots,k\} \quad \text{and} \quad \exists\, j \in \{1,\ldots,k\} : f_j(\mathbf{x}^{(a)}) < f_j(\mathbf{x}^{(b)}),$$
$$\mathbf{x}^{(a)}, \mathbf{x}^{(b)} \in \mathcal{X}, \quad \mathbf{f} = (f_1,\ldots,f_k) : \mathcal{X} \to \mathbb{R}^k, \quad k \in \{n \in \mathbb{Z} \mid n \geq 2\}\,. \tag{2.3}$$

A solution is called *Pareto optimal* if no other feasible solution dominates it. The set of all Pareto-optimal solutions in the decision space is the *Pareto set*, and its image in the objective space is the *Pareto front*. In practice, algorithms compute an approximation of the Pareto front, and the quality of this approximation is assessed using performance indicators such as the hypervolume indicator. [46, 47]

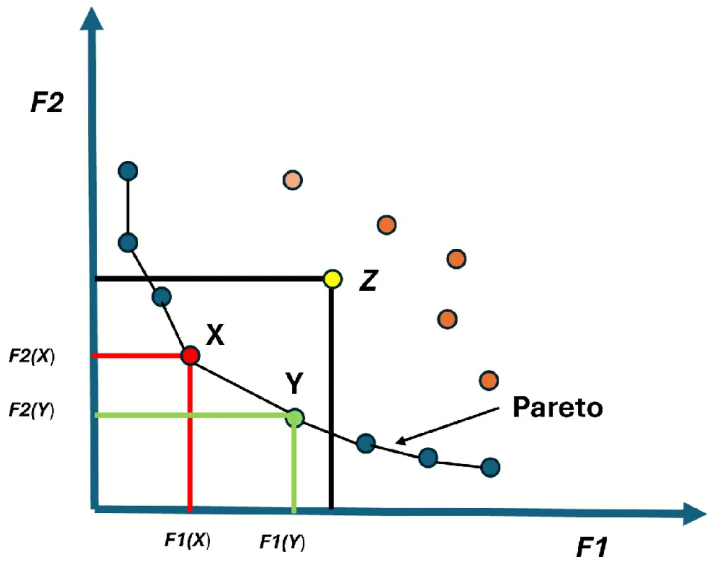


Figure 2.2: Example of a Pareto Front. [1]

Figure 2.2 illustrates the concept of dominated and non-dominated solutions in a multi-objective optimization problem (MOP) with two objectives. The objective is to minimize both objective functions. The connected curve represents the Pareto front (PF), which includes all Pareto-optimal solutions. Solution X_2 has higher values for both f_1 and f_2 when compared to solutions X_1 and Y. Thus, solutions X_1 and Y dominate solution X_2. However, since neither X_1 nor Y dominates the other, they are considered non-dominated solutions [1].

2.3.4 Many-Objective Optimization

Problems with four or more objectives are commonly referred to as many-objective optimization problems [48]. As the number of objectives grows, the proportion of mutually non-dominated solutions in a randomly sampled population increases rapidly, an effect known as dominance resistance. Pareto dominance then loses its ability to discriminate between candidates, so selection pressure towards the Pareto front deteriorates and the search degenerates into an essentially random walk guided mainly by diversity preservation. [48, 49]

The number of solutions required to cover a k-dimensional front grows exponentially with k, visual inspection of the approximation set becomes impractical, and the computation of indicators such as the hypervolume becomes expensive [4]. Algorithm design responds to these issues either by replacing dominance with decomposition into scalar subproblems, by introducing reference points or reference directions to structure the objective space, or by using an indicator as the selection criterion [48, 50].

2.3.5 No Free Lunch Theory

The No Free Lunch (NFL) theorems for optimization state that, averaged over the set of all possible objective functions, all non-repeating search algorithms exhibit identical expected performance [51, 52]. Any advantage that an algorithm demonstrates on one class of problems is therefore compensated by a corresponding disadvantage on another class. Corresponding results have been derived for the multi-objective case. [52, 53]

The practical consequence is that there is no universally superior optimization algorithm and that the question of which algorithm to use is only meaningful with respect to a specific problem class [54]. This provides the methodological justification for the empirical approach taken in this thesis. Rather than seeking a generally best algorithm, the objective is to determine which algorithms perform well on the particular structure of gripper configuration problems, and how sensitive this ranking is to workpiece geometry and to the dimensionality of the objective space [12].

2.3.6 Evolutionary Algorithms

Evolutionary algorithms (EAs) are a family of population-based metaheuristics inspired by the principles of biological evolution [55]. A population of candidate solutions is maintained across discrete generations. At each generation, new candidate solutions are generated

through variation operators, typically crossover, which recombines pairs of existing solutions, and mutation, which introduces random perturbations, and a selection mechanism retains the most promising individuals for the next generation based on their objective values [56]. This loop continues until a termination criterion is met, such as a maximum number of function evaluations or convergence of the population [57].

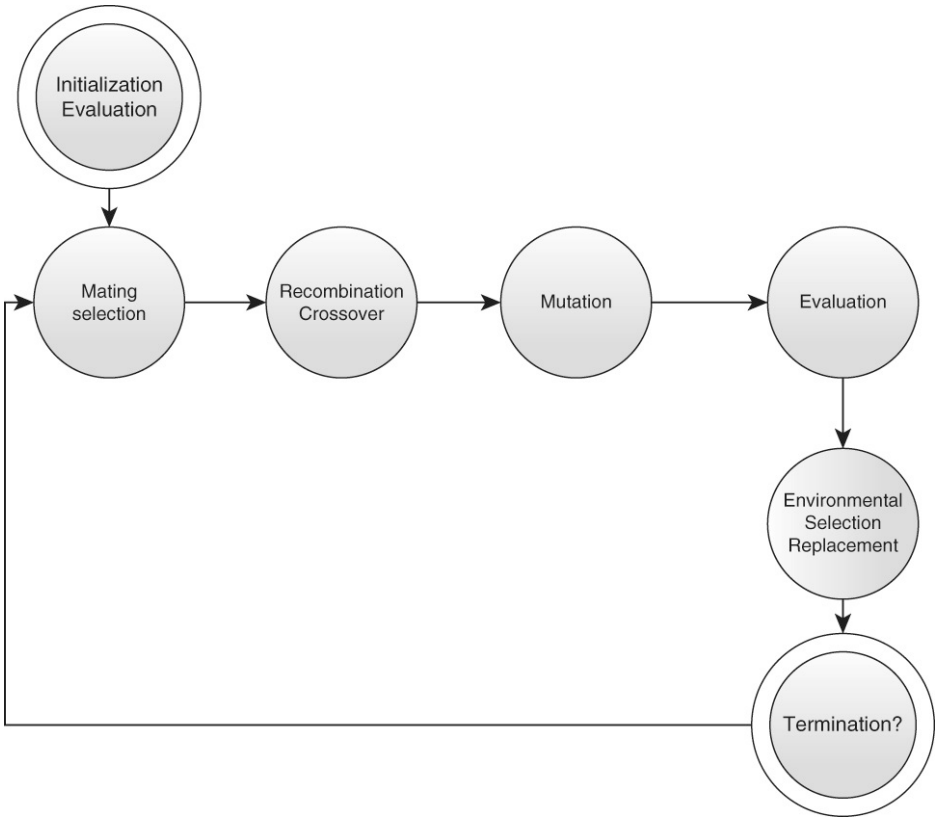


Figure 2.3: The evolutionary cycle, basic working scheme of all EAs [2].

Figure 2.3 shows the basic working scheme shared by all EAs. After initialization and evaluation of the initial population, the algorithm iteratively applies mating selection, recombination (crossover), mutation, evaluation of the offspring, and environmental selection (replacement), until a termination criterion is satisfied. The individual steps of this cycle can be described as follows.

Initialization and Evaluation

The algorithm begins by creating a population of candidate solutions. Random initialization is a simple and commonly used approach, although problem-specific methods can also be employed, for example by including known good solutions as starting points. Each individual is then evaluated to obtain a measure of its quality, referred to as its fitness. When the underlying problem is an optimization problem, this evaluation is performed using the objective function. The evaluated population provides the starting point for the subsequent variation and selection steps. [2]

Mating Selection

Mating selection chooses individuals from the current population to act as parents for offspring generation. Selection introduces a bias toward individuals with better fitness. In genetic algorithms, one possible method is fitness-proportional selection, in which selection probabilities depend on individual fitness values. Another method is tournament selection, in which a small set of individuals is sampled randomly, their fitness values are compared, and the best individual in that set is selected. Thus, in the tournament procedure described here, randomness enters through the choice of participants, while the winner is determined by their fitness. [2]

Recombination (Crossover)

Recombination, also called crossover, generates offspring by combining information from two or more parents. It rearranges information already present in the population and can combine favorable characteristics from different individuals. The specific operator depends on the representation of the candidate solutions. For binary strings, examples include one-point, two-point, and uniform crossover. In one-point crossover, a crossover position is chosen randomly and segments of two parent strings are exchanged. Recombination is used in several evolutionary algorithm families, but it is not a component of evolutionary programming. [2]

Mutation

Mutation is a stochastic variation operator that modifies an individual and introduces new information into the population. Unlike recombination, which rearranges existing information,

mutation can explore regions of the search space not currently covered by the population. Its implementation depends on the representation: binary mutation can flip a randomly chosen bit, whereas evolution strategies can perturb real-valued variables using Gaussian mutations. In evolution strategies, strategy parameters such as the mutation step size influence the operator and can themselves be adapted during the search. [2]

Evaluation of the Offspring

After variation, the offspring are evaluated to determine their fitness before replacement or survivor selection is performed. This follows the same evaluation principle used for the initial population. In applications where fitness evaluation requires a time-consuming numerical model or physical experiment, evaluating candidate solutions can be expensive. [2]

Environmental Selection (Replacement)

Environmental selection determines which individuals survive into the next population. The replacement procedure depends on the algorithm. For example, the evolutionary programming scheme may combine parents and offspring before selecting survivors to keep the population size constant. In contrast, a genetic algorithm may replace the parental population with offspring independently of fitness. For multi-objective optimization, selection requires a ranking mechanism that accounts for several objectives. NSGA-II uses non-dominated ranking and crowding distance, whereas SMS-EMOA uses marginal hypervolume contributions. [2]

Termination

After environmental selection, a termination criterion is checked. If it is not satisfied, the cycle continues with the mating selection of the next generation, otherwise the current population, or the non-dominated subset of all evaluated solutions is returned as the result. [2]

EAs are well suited to black-box multi-objective optimization because they require no gradient information, naturally maintain a diverse population of trade-off solutions, and are robust to noisy or discontinuous objective landscapes [58]. Multi-objective evolutionary algorithms (MOEAs) extend the basic EA framework with selection mechanisms based on

Pareto dominance or decomposition, enabling them to approximate the Pareto front in a single run. [59, 60]

2.3.7 Bayesian Optimization

Bayesian optimization (BO) is a sequential model-based strategy for the global optimization of expensive black-box functions [61]. A probabilistic surrogate model, most commonly a Gaussian process (GP), is fitted to the observations collected so far [62]. The surrogate provides both a mean prediction and an uncertainty estimate at unobserved locations, which together define an *acquisition function* that balances exploration of uncertain regions with exploitation of promising ones [62, 63]. The point that maximizes the acquisition function is selected as the next evaluation, after which the surrogate is updated and the process repeats [64, 3]. Because the surrogate concentrates the sampling effort on informative regions, BO typically requires far fewer function evaluations than population-based methods, which makes it attractive whenever the evaluation budget is the limiting factor [64].

Gaussian Process Surrogates

A Gaussian process defines a distribution over functions and is fully specified by a mean function $m(\mathbf{x})$ and a positive semi-definite covariance or kernel function $\kappa(\mathbf{x}, \mathbf{x}')$, written as $f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), \kappa(\mathbf{x}, \mathbf{x}'))$ [65]. Conditioning the prior on the observed data set $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n_{\text{obs}}}$, where $y_i \in \mathbb{R}$ is the response measured at $\mathbf{x}_i$ and n_{obs} the number of evaluations performed so far, yields a posterior that is again Gaussian, with closed-form predictive mean $\mu(\mathbf{x})$ and non-negative predictive variance $\sigma^2(\mathbf{x})$ at any query point [66]. The kernel encodes the assumed smoothness and length scale of the underlying function, with the Matérn and squared exponential kernels being the most common choices, and its hyperparameters are estimated by maximizing the marginal likelihood [67]. An additional noise term accounts for stochastic or numerically imprecise evaluations [68]. The main limitations of GP surrogates are the cubic scaling of exact inference with the number of observations and a loss of predictive quality in high-dimensional decision spaces [62].

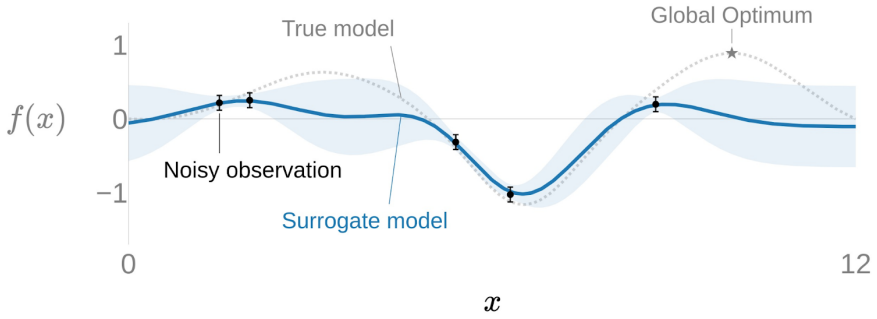


Figure 2.4: Surrogate Model [3].

In Figure 2.4 a surrogate model is fit to five noisy observations using a GP to predict the objective, depicted by the solid line, and uncertainty estimates, illustrated by the width of the shaded bands. This objective is predicted for the entire range of possible parameter values, corresponding to the full x-axis. Importantly, the model is able to predict the outcome and quantify the uncertainty of configurations that have not yet been tested. Intuitively, the uncertainty bands are tight in regions that are well-explored and become wider as we move away from them. [3]

Acquisition Functions

The acquisition function converts the posterior of the surrogate into a scalar utility that quantifies how promising a candidate point is. Expected improvement (EI) measures the expected amount by which a candidate improves on the best observed value,

$$\begin{gathered} \alpha_{\mathrm{EI}}(\mathbf{x}) = \mathbb{E}\big[\max(f_{\min} - f(\mathbf{x}), 0)\big], \\ \mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^d, \quad f_{\min} = \min_{i \in \{1,\ldots,n_{\mathrm{obs}}\}} y_i \in \mathbb{R}, \quad \alpha_{\mathrm{EI}} : \mathcal{X} \to \mathbb{R}_{\geq 0} \end{gathered} \tag{2.4}$$

where $\mathbb{E}[\cdot]$ denotes the expectation taken with respect to the posterior of the surrogate at $\mathbf{x}$ and $f_{\min}$ is the best objective value observed so far. Truncating the improvement at zero renders the acquisition non-negative, so that candidates which are not expected to improve on the incumbent receive zero utility. The expression admits a closed form for a Gaussian posterior. [68, 3]

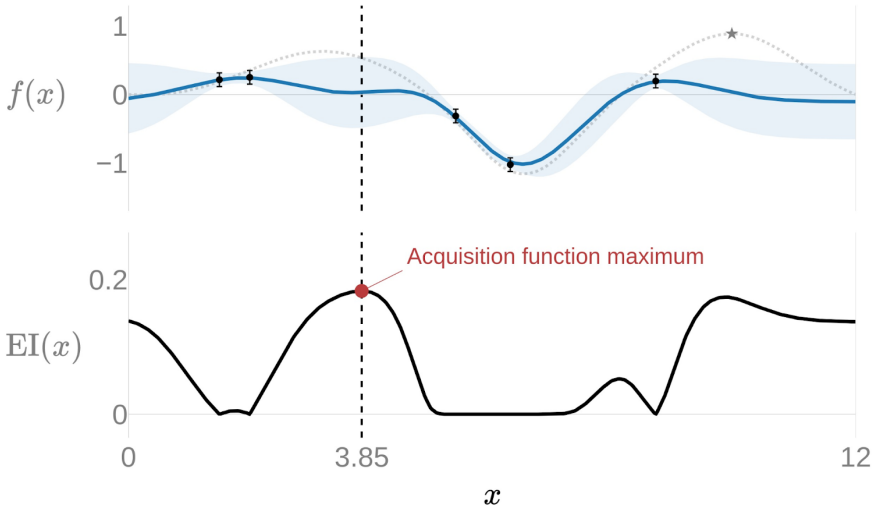


Figure 2.5: Acquisition Function [3].

A visualization of the expected improvement based on the surrogate model predictions is shown in Figure 2.5, where the next suggestion is where the expected improvement is at its maximum. Once a new highest EI is selected and evaluated, the surrogate model is retrained and a new suggestion is made. This process continues iteratively until a stopping condition, set by the user, is reached. [3]

Multi-Objective Bayesian Optimization

Multi-objective Bayesian optimization extends Bayesian optimization to problems with several competing objectives. Surrogate models predict objective values and their uncertainty, while an acquisition function guides the selection of new candidates. The aim is to approximate a set of Pareto-optimal trade-offs rather than identify a single optimum [69].

Different acquisition strategies address this task in different ways. Scalarization combines the objectives into a scalar utility and can explore different trade-off regions by varying the scalarization parameters. Hypervolume-based methods assess candidates according to their expected contribution to the dominated objective-space volume [69]. Information-theoretic methods instead seek candidates expected to provide information about the Pareto front [70].

Before model-based candidate selection, initial observations can be obtained using a space-filling experimental design [62]. For bounded continuous variables, one possible starting

point is the center of the search space, given by the midpoint of each variable's interval. Additional points can be generated using a Sobol sequence. Sobol sequences are low-discrepancy sequences designed to distribute points across a bounded space. [71]

A Gaussian process can be used to model an individual objective, and separate models can be fitted to multiple objectives [69]. The covariance kernel specifies how function values at different inputs are related. For example, the radial basis function (RBF) kernel assigns greater correlation to nearby points. Its length-scale parameters control how quickly that correlation decreases with distance, expressing an assumption of a smooth objective function [62].

Expected hypervolume improvement measures the expected increase in dominated hypervolume relative to a reference point. Its noisy, batch-capable extension, qNEHVI (q-Noisy Expected Hypervolume Improvement), accounts for uncertainty about the objective values at previously evaluated points and can assess a set of candidates jointly [69]. The related qLogNEHVI (q-Log Noisy Expected Hypervolume Improvement) formulation uses logarithmic and smooth reformulations to improve numerical behavior when acquisition values or gradients become very small [72].

Multi-objective Bayesian optimization is attractive when evaluations are expensive and the evaluation budget is limited. However, fitting surrogates and optimizing acquisition functions also incur computational cost, which can increase with the number of observations, objectives, and candidates considered in a batch [69, 72]. In particular, exact Gaussian process inference has a computational cost of $\mathcal{O}(n_{\text{obs}}^3)$, where n_{obs} is the number of observations used to fit the model [73]. This can limit scalability as observations accumulate.

2.3.8 Performance Indicators

Because a multi-objective optimizer returns a set rather than a single point, comparing algorithms requires indicators that map an approximation set to a scalar quality value [4]. Indicators are usually grouped according to whether they assess convergence towards the front, the diversity of the solutions along the front, or both properties at once.

Hypervolume

The *hypervolume* indicator (HV) measures the volume of the objective space that is dominated by the approximation set and bounded by a reference point $\mathbf{z}_{\text{ref}}$,

$$
\begin{aligned}
&\mathrm{HV}(A, \mathbf{z}_{\mathrm{ref}}) = \lambda\Big(\bigcup_{\mathbf{a}\in A} [\mathbf{f}(\mathbf{a}), \mathbf{z}_{\mathrm{ref}}]\Big), \\
&A \subset \mathcal{X}, \quad |A| < \infty, \quad \mathbf{f} = (f_1, \ldots, f_k) : \mathcal{X} \to \mathbb{R}^k, \quad k \in \{n \in \mathbb{Z} \mid n \geq 1\}, \\
&\mathbf{z}_{\mathrm{ref}} = (z_{\mathrm{ref},1}, \ldots, z_{\mathrm{ref},k}) \in \mathbb{R}^k, \quad f_i(\mathbf{a}) \leq z_{\mathrm{ref},i}, \quad \forall\, \mathbf{a} \in A, \quad \forall\, i \in \{1, \ldots, k\}, \\
&\lambda : \mathcal{B}(\mathbb{R}^k) \to \{t \in \mathbb{R} \mid t \geq 0\} \cup \{\infty\}, \quad \mathrm{HV}(A, \mathbf{z}_{\mathrm{ref}}) \in \{t \in \mathbb{R} \mid t \geq 0\}.
\end{aligned}
\tag{2.5}
$$

where λ denotes the Lebesgue measure on the Borel sets $\mathcal{B}(\mathbb{R}^k)$ and A is the finite approximation set returned by the optimizer [46, 74]. The reference point $\mathbf{z}_{\mathrm{ref}}$ must be dominated by every member of A, that is $f_i(\mathbf{a}) \leq z_{\mathrm{ref},i}$ for all $\mathbf{a} \in A$ and all i, so that each box $[\mathbf{f}(\mathbf{a}), \mathbf{z}_{\mathrm{ref}}]$ is non-empty and the indicator is well defined and non-negative. The hypervolume is the most widely used indicator because it is Pareto compliant, meaning that a strictly dominating set never receives a worse value, and because it requires no knowledge of the true Pareto front [4, 46].

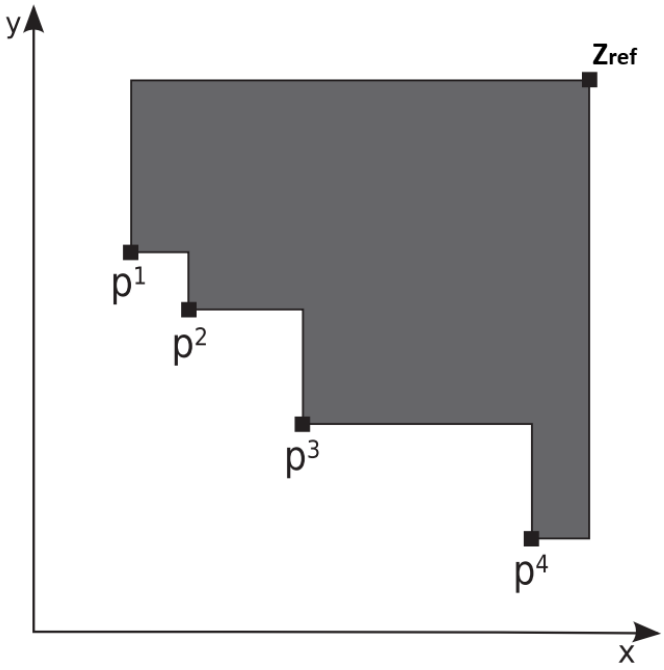


Figure 2.6: Hypervolume indicator [4].

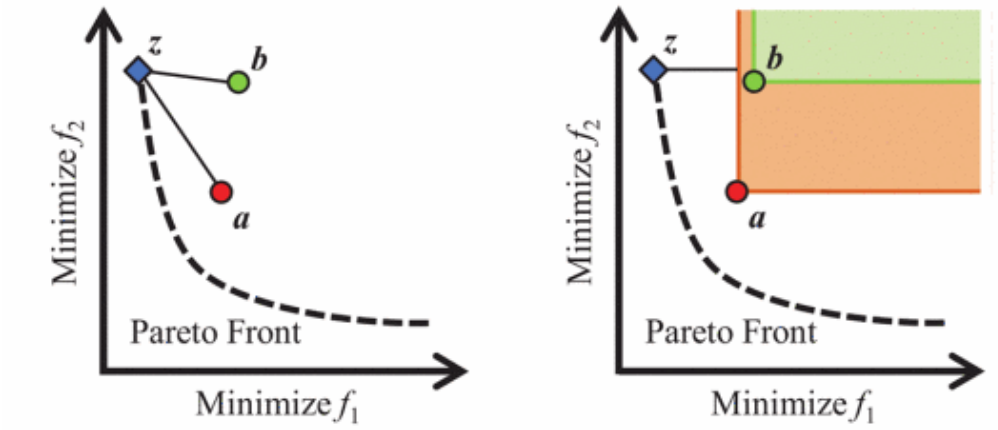
Figure 2.6 shows a two-dimensional example of the hypervolume (an area) and a fixed ref-

erence point $\mathbf{z}_{\text{ref}}$.

Inverted generational distance.

The *inverted generational distance* (IGD) measures the mean distance from each point of a reference set Z to the closest point of the approximation set and therefore requires a reference front, which is normally unavailable for real-world black-box problems [74]. IGD is furthermore not Pareto compliant: a set that dominates another can receive a worse indicator value [74]. The modified variant IGD$^+$ removes this defect by replacing the Euclidean distance with a dominance-aware distance that only penalizes those objective components in which the approximation set is actually worse than the reference point [75].

Figure 2.7 illustrates the difference between the distance calculations for a two-objective minimization problem. Although solution **a** dominates solution **b**, its Euclidean distance from the reference point **z** is larger. Thus, IGD can prefer **b** over **a**. In contrast, IGD$^+$ measures the distance from **z** to the region dominated by each solution. Consequently, the distance for **a** is no greater than that for **b**.



(a) Distance calculation in the IGD. (b) Distance calculation in the IGD$^+$.

Figure 2.7: Distance calculation for IGD and IGD$^+$ in a two-objective minimization problem. Reproduced from Fig. 3 of [5].

$$\mathrm{IGD}^{+}(A,Z) = \frac{1}{|Z|}\sum_{\mathbf{z}\in Z}\min_{\mathbf{a}\in A}\sqrt{\sum_{i=1}^{k}\left(\max\{\,a_i - z_i, 0\,\}\right)^2}, \qquad \mathrm{IGD}^{+}(A,Z) \in \{r \in \mathbb{R} \mid r \geq 0\},$$
$$A, Z \subset \mathbb{R}^k, \quad 0 < |A| < \infty, \quad 0 < |Z| < \infty, \quad k \in \{n \in \mathbb{Z} \mid n \geq 1\},$$
$$\mathbf{a} = (a_1, \ldots, a_k), \quad \mathbf{z} = (z_1, \ldots, z_k), \quad \forall\, \mathbf{a} \in A, \quad \forall\, \mathbf{z} \in Z. \tag{2.6}$$

In Equation (2.6), A denotes the approximation set under evaluation, while a_i and z_i denote the i-th objective values of an approximation point and a reference point, respectively. Lower values indicate better approximation quality, and IGD^{+} is weakly Pareto compliant [75]. Because the true Pareto front of the gripper design problem is unknown, the reference set Z is constructed as the non-dominated union of the fronts of all compared runs [74]. IGD^{+} is therefore a *relative* measure: it quantifies how well an individual run approximates the best front jointly attainable by the whole set of runs, and it is optimistically biased in favor of those runs that contribute many points to Z [74].

2.3.9 Reference-Direction Generation

Reference directions specify target regions in objective space and are used by algorithms such as NSGA-III and MOEA/D to distribute their search. Two generation methods available in `pymoo` are Das–Dennis and energy-based generation [76]. The Das–Dennis method constructs a regular grid on the unit simplex. For m objectives and p partitions, the number of directions N is

$$N = \binom{p+m-1}{m-1}, \qquad p, m \in \{n \in \mathbb{Z} \mid n \geq 1\}, \quad N \in \{n \in \mathbb{Z} \mid n \geq 1\}. \tag{2.7}$$

This construction is deterministic, but the direction count grows combinatorially with the objective count for a fixed partition setting. It also restricts the available direction counts to those produced by the grid construction. [76, 50]

Energy-based generation distributes a prescribed number of points on the unit simplex by minimizing a repulsive energy function. Unlike Das–Dennis, it permits direct specification of the number of directions, making it useful when a fixed reference-set size is required in higher-dimensional objective spaces. [76, 77]

2.4 Multi-Objective Evolutionary Algorithms

The candidate algorithms introduced in this section are all members of the multi-objective evolutionary algorithm (MOEA) family. Each follows the general EA loop described in Section 2.3.6 but differs in how it guides selection toward a well-distributed approximation of the Pareto front. Brief introductions to each algorithm are given below.

2.4.1 NSGA-II

The Non-dominated Sorting Genetic Algorithm II (NSGA-II) is among the most widely adopted MOEAs. Its selection mechanism is based on two criteria applied in sequence. First, the entire population is sorted into non-domination ranks, so that all non-dominated solutions receive rank 1, they are removed, and the process repeats until every individual has a rank. Second, within the same rank, solutions are ordered by *crowding distance*, a density estimate that measures how isolated a solution is from its neighbors in the objective space. Solutions with a higher crowding distance are preferred, encouraging spread along the Pareto front. NSGA-II uses an elitist strategy in which the combined parent and offspring population of size $2N$ is truncated to N using these two criteria. Its computational complexity per generation is $\mathcal{O}(kN^2)$, where k is the number of objectives and N is the population size. [78]

2.4.2 NSGA-III

NSGA-III extends NSGA-II to problems with four or more objectives (many-objective problems), where crowding distance becomes an unreliable diversity measure. Instead of crowding distance, NSGA-III uses a set of structured *reference points* distributed uniformly on a normalized hyperplane in the objective space. Solutions are associated with their closest reference point, and selection preference is given to solutions that are associated with under-represented reference points and lie closest to their reference line. This mechanism explicitly promotes diversity across the objective space and has been shown to outperform NSGA-II on problems with four or more objectives. [79]

2.4.3 MOEA/D

MOEA/D (Multi-Objective Evolutionary Algorithm based on Decomposition) takes a fundamentally different approach from Pareto-ranking methods. The multi-objective problem is decomposed into N scalar subproblems using a set of evenly distributed weight vectors and an aggregation function such as the weighted Chebyshev scalarization. Each subproblem is optimized simultaneously, with solutions sharing information within a defined neighborhood of weight vectors. This neighborhood-based collaborative search makes MOEA/D computationally efficient and well suited to problems where the Pareto front has a regular structure. Its complexity per generation is lower than NSGA-II, as it avoids global non-dominated sorting. [80, 76]

2.4.4 SMS-EMOA

The S-Metric Selection Evolutionary Multi-Objective Algorithm (SMS-EMOA) uses the hypervolume indicator directly as its selection criterion. It operates as a steady-state algorithm, so that at each step a single offspring is generated, added to the population, and the individual that contributes the least to the population hypervolume is removed. This ensures that the population converges toward solutions that maximize the dominated hypervolume with respect to a reference point. SMS-EMOA is particularly well suited to problems where solution quality is to be measured by the hypervolume indicator, as its selection pressure is directly aligned with that metric. [81]

2.4.5 CMOPSO

The Competitive Multi-Objective Particle Swarm Optimizer (CMOPSO) adapts the particle swarm optimization (PSO) framework to the multi-objective setting. In standard PSO, each particle moves through the decision space guided by its personal best position and a global best position. CMOPSO introduces a competitive mechanism in which particles compete in pairs within the swarm, and the losing particle updates its velocity by learning from the winning particle rather than from a fixed global best. An external archive of non-dominated solutions is maintained to provide guidance and preserve diversity across the Pareto front. This competitive selection strategy avoids premature convergence to a single region of the Pareto front and has been shown to achieve fast convergence while remaining competitive with dominance-based MOEAs on standard benchmark problems. [82]

3 Literature Review

This section compares previous studies according to their optimization target, the gripper properties treated as fixed, and their relevance to task-specific gripper configuration. The reviewed literature can be divided into four groups: grasp planning with fixed end-effectors, multi-suction grasping, two-finger gripper design, and comparative studies of multi-objective optimization algorithms.

3.1 Grasp Planning with Fixed End Effectors

Most robotic grasping research treats the end-effector geometry as a fixed input and searches for a feasible grasp pose on the workpiece. Zurbrügg et al. [83], for example, optimize grasp selection while assuming that the geometry and contact configuration of the gripper have already been defined. Analytical grasp-planning methods evaluate candidate contacts using geometric, force-closure, or form-closure conditions, whereas data-driven methods estimate grasp quality from training data. Despite their methodological differences, both approaches primarily vary the grasp pose rather than the gripper geometry.

Although Todescato et al. [6] establish gripper geometry as an optimization variable, their formulation is limited to a single suction cup. In their objective-function evaluation the suction-contact feasibility is assessed using quasi-static criteria related to the methods presented by Kleeberger et al. [84] and the compliant suction-contact model of Mahler et al. [85]. However, their formulation primarily considers the availability of valid suction grasps and does not explicitly evaluate the distribution/coverage of the resulting grasp set. Building on this work, the present thesis explores different approaches to evaluating multi-suction-cup layouts. In addition to grasp availability, these include measures of the spatial and orientational distribution of valid grasps. These metrics distinguish between layouts that provide a similar number of valid grasps but differ in how broadly those grasps are distributed across positions and orientations.

The studies therefore address complementary but different questions. Grasp planning determines how an existing gripper should interact with a workpiece, whereas gripper-configuration optimization determines which geometry should be available before grasp planning is performed. Improving the grasp planner cannot fully compensate for a suction layout that cannot establish sufficient seals or for a jaw contour that cannot reach suitable contact regions. Consequently, fixed-gripper grasp-planning results do not directly resolve the end-effector design problem considered in this thesis.

3.2 Comparison of Multi-Suction Grasping Studies

Existing studies on multi-suction grippers address grasp prediction, experimental stability assessment, and load-aware motion planning, but generally assume predefined suction-cup layouts. Schillinger et al. [86] predict grasp quality from RGB-D images and optimize the grasp pose and gripper selection from a predefined set of multi-suction grippers. Their method selects among existing layouts rather than optimizing the positions of the individual cups.

Geng et al. [87] experimentally investigate how suction-cup size, geometric arrangement, center of gravity offset, and robot acceleration influence grasp stability. Although different arrangements are evaluated, these are predefined experimental configurations rather than layouts generated through multi-objective optimization.

Lee et al. [88] model the load distribution among multiple suction cups using a quadratic program. They incorporate the resulting grasp-failure constraints into time-optimal trajectory planning to support the fast and reliable manipulation of heavy objects. Their work focuses on load prediction and motion planning for an existing gripper configuration, not on optimizing the cup arrangement.

These studies therefore address complementary aspects of multi-suction grasping. Schillinger et al. focus on grasp-pose and gripper selection, Geng et al. examine the effects of gripper configuration and handling conditions experimentally, and Lee et al. model load distribution during dynamic manipulation. However, none of these studies formulates the positions of the individual suction cups as decision variables in a multi-objective layout-optimization problem.

3.3 Comparison of Two-Finger Gripper Studies

Research on two-finger gripper design includes the optimization of compliant fingers, underactuated mechanisms, and task-specific jaw geometries. Sun et al. [89] use three-dimensional topology optimization to design compliant continuum fingers and a lightweight gripper base. The resulting fingers deform passively to adapt to objects with different shapes. Similarly, Xie et al. [90] optimize the topology of flexible fingers while accounting for contact behavior and contact-stress constraints. Their objective is to achieve adaptive, damage-free grasping of fragile and geometrically variable objects. Both studies focus on structural compliance and deformation, whereas this thesis considers the geometric optimization of rigid jaw contours.

Dong et al. [91] optimize the geometry and transmission system of an underactuated tendon-driven gripper. Their design variables include phalange dimensions, palm width, joint-mandrel radii, and tendon-routing parameters. The optimization aims to improve force transmission and grasp stability. Although this approach optimizes gripper geometry, it concerns the dimensions and actuation mechanism of an articulated gripper rather than the contact contour of a rigid parallel jaw.

Reichert [92] developed the direct predecessor of the parallel-jaw optimization framework continued in this thesis. The work represents the contact contour of a rigid parallel jaw using a parametric spline and evaluates candidate contours through a collision-map-based procedure. Intersections close to the intended workpiece boundary contribute positively to a geometric key performance indicator (KPI), whereas intersections extending into the workpiece interior are penalized. The KPI therefore measures the geometric conformity of the generated jaw contour at a specified evaluation configuration rather than providing a complete force- or wrench-based assessment of grasp stability.

Reichert formulates the jaw-design problem using two fixed workpiece orientations, each contributing an objective to the optimization. This demonstrates the feasibility of combining spline-based jaw generation, geometric KPI evaluation, and multi-objective black-box optimization. However, the formulation is restricted to the two investigated evaluation configurations. This thesis continues Reichert's work by extending the evaluation from two fixed configurations to a set of n_{or} predefined grasp points.

The optimization therefore seeks Pareto-optimal rigid jaw contours that provide different compromises in geometric conformity across the predefined grasp points. Compared with

the studies on compliant and underactuated grippers, the formulation concerns rigid task-specific contact contours. Compared with Reichert's direct predecessor formulation, the principal extension is the evaluation and optimization of one separate objective for each of the n_{or} predefined grasp points.

3.4 Comparative Studies of Multi-Objective Optimization Algorithms

Comparative studies show that optimizer performance depends strongly on the problem and experimental setup. Tanabe and Ishibuchi [93] evaluate evolutionary algorithms on a suite of real-world multi-objective problems with varied characteristics, while Tian et al. [94] propose selecting algorithms based on problem features. Ishibuchi et al. [95] further demonstrate that termination conditions, population sizes, performance indicators, and test problems can change algorithm rankings. Audet et al. [46] emphasize that indicators assess different properties of Pareto-front approximations, such as convergence, distribution, and spread.

For expensive optimization, Qin et al. [96] compare two combinations of evolutionary and Bayesian optimization on ZDT (Zitzler-Deb-Thiele) and DTLZ (Deb-Thiele-Laumanns-Zitzler) problems and find that Bayesian evolutionary optimization is more efficient under their experimental conditions. Santoni et al. [16] compare high-dimensional Bayesian optimization methods on single-objective BBOB (Black-Box Optimization Benchmarking) functions and report that performance varies with dimensionality, function landscape, and evaluation budget. These findings are therefore specific to the investigated problems and settings.

Such results cannot be transferred directly to the collision-aware suction-layout and spline-based jaw-contour problems considered in this thesis. Therefore, this thesis compares selected multi-objective evolutionary and Bayesian algorithms using defined evaluation budgets, initialization conditions, and repeated runs where available. The aim of this comparison is to identify methods suitable for these specific gripper-design problems rather than a universally superior optimizer.

3.5 Comparative Synthesis and Research Gap

The reviewed literature reveals three related research gaps. First, grasp-planning studies generally optimize grasp poses for predefined end-effectors rather than the gripper configuration itself. Second, existing multi-suction studies generally assume predefined cup layouts. For parallel-jaw design, Reichert [92] provides the spline-based formulation used in this thesis but considers only two fixed evaluation configurations. Third, existing algorithm comparisons do not determine which optimization methods are best suited to the two geometric gripper-design problems considered in this thesis.

For the multi-suction gripper, the unresolved problem is to optimize the positions of the individual cups to maximize grasp coverage on every workpiece and the area formed by the cup arrangement. For the two-finger gripper, this thesis continues Reichert's framework by optimizing a rigid spline-based jaw contour across predefined grasp points. Each grasp point contributes an individual geometric conformity objective. Both formulations are expensive multi-objective black-box problems, but their different representations, number of objectives, and evaluation procedures may favor different optimization methods.

Within the literature reviewed for this thesis, no controlled comparison of evolutionary and Bayesian multi-objective algorithms was identified for these two gripper design formulations.

3.6 Research Positioning and Contribution

This thesis addresses these gaps by treating gripper geometry as the primary optimization variable. The multi-suction formulation optimizes the spatial arrangement of individual suction cups, whereas the two-finger formulation optimizes spline-based contours for rigid parallel jaws. Candidate designs are assessed using problem-specific geometric and collision-aware evaluations.

A modular framework separates geometric evaluation from optimization through a common interface. This allows the applicable algorithms to operate on the same problem formulation, constraints, and evaluation conditions. Selected evolutionary and Bayesian multi-objective algorithms are compared under defined evaluation budgets using repeated independent runs, convergence histories, and multi-objective performance indicators. Evaluation counts and optimizer overhead are reported separately where computational cost is relevant.

The contribution of this thesis is therefore a problem-specific comparison rather than a claim of universal algorithm superiority. The results provide evidence for selecting optimization methods for the suction-layout and rigid-jaw-contour formulations considered here.

4 Methodology

This chapter presents the methodological foundations of the gripper optimization study. It describes the research design, problem formulation, geometric parameterizations, objective criteria, selected optimization frameworks and algorithms, workpiece selection, evaluation architecture, and performance indicators. Implementation details are presented in Chapter 5, and the experimental settings in Chapter 6.

4.1 Research Design

4.1.1 Research Paradigm

In this thesis, gripper configurations are represented by parameterized mathematical models and assessed using computational geometry. The optimization methods access these assessments as black-box evaluations. This approach reflects the dependence of gripper quality on the geometric interaction between the gripper and the workpiece, which can be difficult to characterize analytically. The selected optimization methods do not require gradients of the black-box objective functions.

4.1.2 Study Structure

The suction-cup and parallel-jaw grippers are treated as separate optimization studies. Each gripper–workpiece combination defines a separate problem instance, and each optimization run considers exactly one workpiece.

For the suction-cup gripper, the positions of three cups are optimized with respect to the spatial and orientational diversity of valid grasps, the number of distinct valid grasps, and the triangle area formed by the cups.

For the parallel-jaw gripper, a spline-based jaw contour is optimized for a predefined grasp set on the selected workpiece. The contour is evaluated according to its geometric proximity to the workpiece while excluding contact conditions classified as infeasible by the evaluator.

Results from different workpieces or gripper types are analyzed separately and are not combined into a common Pareto front.

4.1.3 Experimental Setup

The selected algorithms are compared within each gripper–workpiece instance. To reduce differences unrelated to the optimization methods, they use the same geometric parameterization, variable bounds, objective definitions, feasibility conditions, and evaluation service.

Independent repetitions of the stochastic methods are used to assess performance variability and robustness. Evaluation budgets are identical within each optimizer family but differ between evolutionary and Bayesian optimization, as specified in Section 5.1.

4.2 Problem Formulation

4.2.1 Parameterization of Gripper Geometry

A central methodological decision is the representation of gripper geometry as a finite-dimensional decision vector. The representation must provide sufficient geometric flexibility while remaining compact, interpretable, and suitable for black-box optimization [97, 6, 98].

Suction-Cup Gripper Representation

The suction-cup gripper consists of three cups whose positions must be represented in Cartesian space for grasp generation, collision checking, and objective computation. Independently optimizing their planar Cartesian coordinates would require six decision variables. A reduced polar parameterization is therefore selected to constrain the search space and encode geometric structure.

The cup centers lie on a common circle of radius r, with their positions determined by the angles α_i. The resulting decision vector is

$$\begin{aligned} \mathbf{q}_{\mathrm{SC}} &= (r,\alpha_1,\alpha_2,\alpha_3)\,, \qquad \mathbf{q}_{\mathrm{SC}} \in \mathbb{R}^4, \\ r \in \mathbb{R}, \qquad \alpha_i \in \mathbb{R}, \quad \forall i &\in \{1,2,3\}. \end{aligned} \tag{4.1}$$

which reduces the number of optimization variables from six to four.

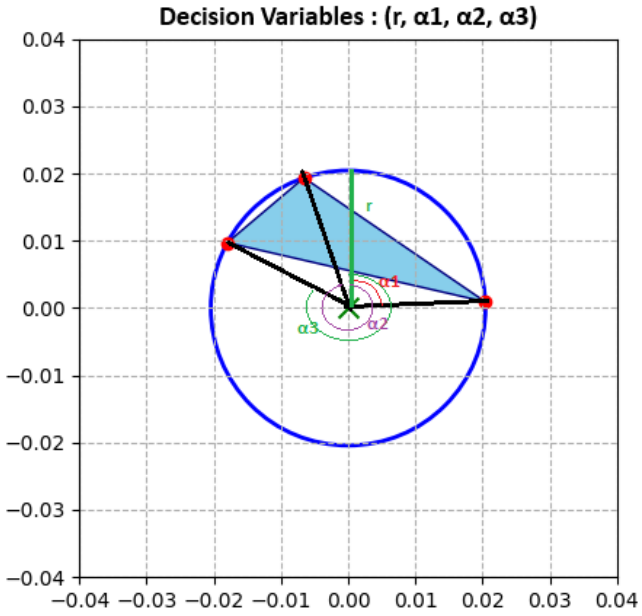


Figure 4.1: Polar parameterization of the suction-cup gripper.

Figure 4.1 illustrates this parameterization. The common radius r determines the radial distance of each cup center from the gripper Tool Center Point (TCP), while the angles α_i determine the relative cup arrangement and its overall in-plane orientation. For this thesis, the gripper height is fixed at $z_0 = 0.326$ m, adopted from the suction-cup configuration employed in the bin-picking demonstration conducted by Fraunhofer IPA. The Cartesian

position of cup i is obtained through the polar-to-Cartesian transformation

$$\mathbf{p}_i = \begin{bmatrix} r\cos(\alpha_i) \\ r\sin(\alpha_i) \\ z_0 \end{bmatrix}, \qquad \forall i \in \{1,2,3\}, \\ r,z_0 \in \mathbb{R}, \qquad \alpha_i \in \mathbb{R}, \quad \mathbf{p}_i \in \mathbb{R}^3, \qquad \forall i \in \{1,2,3\}. \tag{4.2}$$

The resulting Cartesian gripper configuration is

$$\mathbf{P} = [\mathbf{p}_1,\mathbf{p}_2,\mathbf{p}_3], \qquad \mathbf{P} \in \mathbb{R}^{3\times 3}. \tag{4.3}$$

Thus, optimization operates in the reduced four-dimensional parameter space of $\mathbf{q}$, while grasp evaluation operates on the derived Cartesian configuration $\mathbf{P}$.

All three angles are retained because grasp evaluation does not separately explore rotations of the cup configuration about the local z-axis, which is normal to the gripper plane. Changing all three angles by the same amount rotates the configuration about the gripper origin without altering the relative cup arrangement. Fixing $\alpha_1 = 0$ would instead constrain the first cup to the positive direction of the grasp frame's first in-plane axis. Although the remaining angles could describe different relative arrangements, their overall orientation within that frame would be restricted. Retaining all three angles therefore allows the optimizer to explore both cup arrangement and in-plane orientation within the prescribed angular bounds.

The common-circle constraint limits the search space while retaining flexibility for diverse triangular cup arrangements. A spatially distributed triangular arrangement is adopted as a design assumption to favor load distribution and resistance to external disturbances [34]. The triangle area serves as a geometric proxy for this distribution, not as a guarantee of grasp stability, which also depends on seal quality, available suction forces, workpiece properties, and external loading.

Parallel-Jaw Gripper and Contour Representation

The parallel-jaw (PJ) gripper applies gripping forces through two opposing jaw surfaces. Their contact contours determine how closely the gripper conforms to the workpiece at the predefined grasp orientations. To represent smooth contour variations with a compact design vector, the jaw contour is defined by a cubic interpolating spline [11, 99, 100].

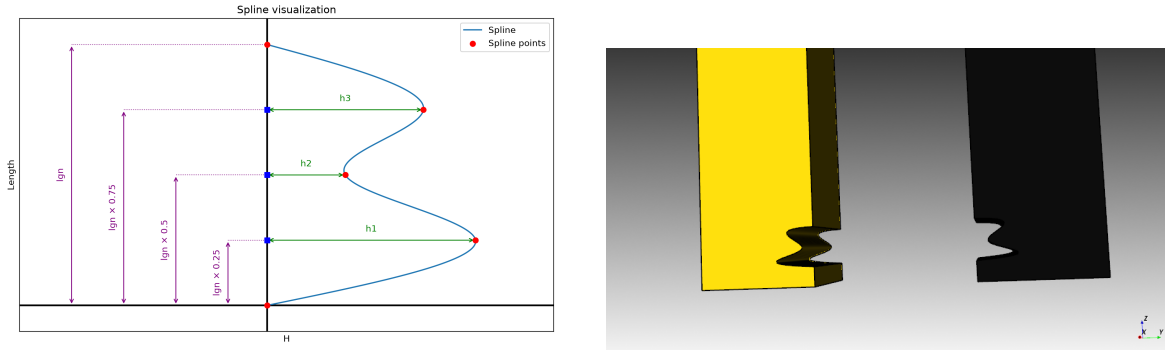
The contour lies in a plane spanned by the jaw depth and jaw length directions. It is parameterized by the contour length ℓ_{gn} and three depth offsets h_1, h_2, and h_3. These define five interpolation points: the endpoints lie at zero depth, while the three interior points are positioned at quarter intervals along the contour length. The contour definition is

$$\mathbf{q}_{\text{PJ}} = \begin{bmatrix} \ell_{\text{gn}} & h_1 & h_2 & h_3 \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^4. \tag{4.4}$$

The points on the cubic spline are defined as

$$\begin{aligned} &\mathbf{s}_0 = (0,0), \qquad \mathbf{s}_4 = (0,\ell_{\text{gn}}), \\ &\mathbf{s}_j = \left(h_j, \frac{j}{4}\ell_{\text{gn}}\right) \in \mathbb{R}^2, \qquad \forall j \in \{1,2,3\}. \end{aligned} \tag{4.5}$$

Unlike B-spline control points, the interpolation points lie on the resulting curve. The endpoints at zero depth make the contour meet the planar jaw face at both ends, where the three interior offsets determine the contour shape. Using four design variables provides a compact way to vary the jaw profile while maintaining a smooth interpolating curve.



(a) (b)

Figure 4.2: (a) Cubic spline interpolating the jaw-contour points. (b) Parallel-jaw gripper with the parameterized contours on its opposing jaw surfaces.

Figure 4.2(a) shows the interpolating spline, and Figure 4.2(b) shows the corresponding gripper geometry. To construct the jaw, the contour is closed into a planar profile using straight segments to a fixed cutting depth, then extruded through the jaw depth to remove material from the basic jaw body. Geometry is generated with CadQuery [101] and exported in the mesh representation used by the evaluation pipeline. Only ℓ_{gn} and the three offsets vary between candidates, while all other jaw dimensions remain fixed.

Each candidate jaw geometry is generated once per evaluation request. For each predefined workpiece orientation, the evaluator then voxelizes the gripper and the rotated workpiece for the corresponding collision and proximity calculations.

4.2.2 Selection of Objective Functions for Multi-Suction Cup Gripper

The objective functions define the properties used to distinguish effective gripper configurations. They must represent meaningful geometric performance criteria while remaining computable through the available evaluation models [6, 102].

For the multi-suction cup gripper, three objectives are used.

Grasp diversity

The first objective is the diversity of the valid grasps in position and orientation. A layout that can grasp the workpiece in only one region is of limited use when the part arrives in varying poses or when parts of its surface are blocked by fixtures or neighboring objects. A diversity measure captures this operational flexibility [103, 104].

Number of distinct grasps

Diversity alone does not show how many alternatives exist, because two well-separated grasps can already produce a large spread. The number of distinct valid grasps is therefore the second objective, as a measure of robustness: the more distinct grasps a layout offers, the more likely a usable one remains when others are excluded by the scene or by reachability. Only distinct grasps are counted, so that near-identical grasps cannot inflate the score.

Support area

The third objective is the area of the triangle spanned by the three cups. A wider support polygon gives a longer lever arm against tilting moments, so the area serves as a stability proxy [33]. It is a purely geometric measure. The area objective itself does not model suction force, sealing quality, material properties, or the center of mass. The area was also chosen because it conflicts with the first two objectives. A wide layout needs larger surface regions on which all three cups can seal, which reduces both the number and the spread of feasible

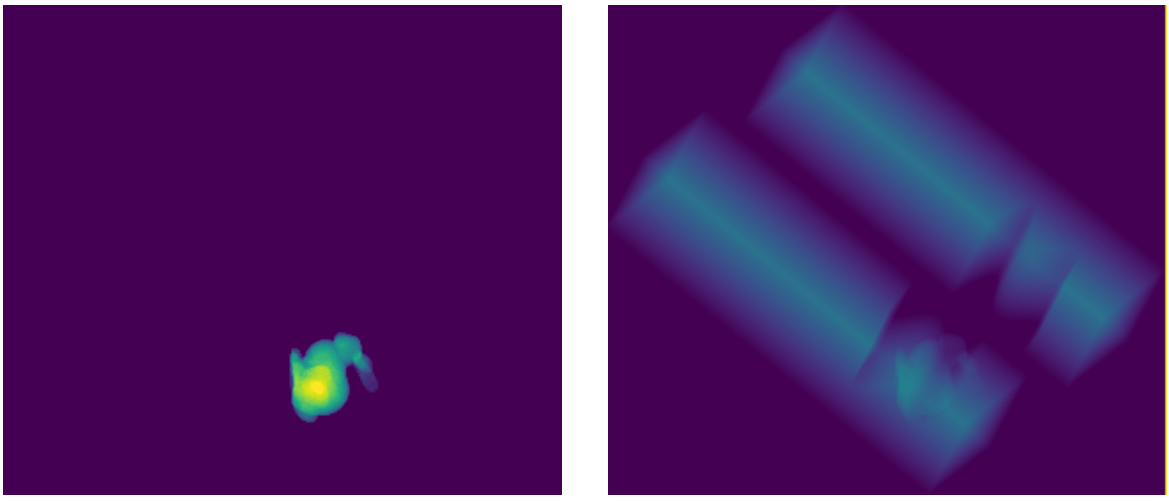
grasps. This conflict is the central trade-off of the suction-cup design and is why a multi-objective formulation is needed. All three objectives are normalized so that they remain comparable across workpieces and within the hypervolume indicator.

4.2.3 Selection of Objective Functions for Two-Finger Gripper

For the two-finger gripper, the objectives measure the geometric proximity between the workpiece and the two finger contours. This criterion is evaluated for a fixed set of $n_{\text{or}} = 4$ predefined workpiece orientations. The method is inspired by Snuggle-Pack, which uses voxel-based collision and proximity analysis to evaluate how closely an object fits its surroundings [105]. Here, that general principle is adapted to gripper evaluation where the workpiece is assessed relative to the two jaws, rather than relative to objects and walls in a packing container.

Geometric proximity score.

For each gripper design and workpiece orientation, the evaluator uses voxel grids to calculate jaw-specific overlap and proximity maps. Indices with high combined proximity scores are favored, while indices classified as colliding by the implementation are excluded. The score is retained only if the selected jaw indices satisfy the implementation's opening condition.



(a) Shifted workpiece (b) Combined occupancy

Figure 4.3: Example voxel projections.

Figure 4.3 shows diagnostic projections of the voxel grids. The workpiece grid is shifted using the selected left-jaw index, and the combined image shows its union with the gripper in its modeled pose. Both projections are summed along the gripper length axis. These images do not show both jaws moved to their selected positions and do not independently establish grasp feasibility.

The score is a geometric proxy and does not directly measure contact force, friction, material behavior, or grasp stability. As discussed in Section 5.3.2, the FFT calculation uses cyclic convolution. Its collision-map interpretation therefore differs from a direct collision test of the workpiece placement shown in the diagnostic images.

4.3 Optimization Library Selection

The evolutionary optimization algorithms are selected from the `pymoo` framework. `pymoo` provides a unified Python interface for multi-objective evolutionary optimization, constraint handling, performance evaluation, and visualization. Using algorithms from a common framework reduces implementation differences and improves the reproducibility of the comparison. The framework also provides established multi-objective performance indicators, including hypervolume and inverted generational distance plus. [76]

Bayesian multi-objective optimization is represented using the `Ax` platform and its underlying `BoTorch` functionality. Bayesian optimization constructs probabilistic surrogate models of the black-box objective functions and uses an acquisition function to select promising candidate evaluations. `Ax` provides the experiment-management interface, while `BoTorch` provides the probabilistic modeling and acquisition-function capabilities. [3, 106]

The use of established open-source frameworks avoids the need to implement the optimization methods, surrogate models, and acquisition functions from scratch. It also supports a more reproducible comparison between evolutionary and Bayesian optimization strategies.

4.4 Algorithm Selection

For the experimental comparison, NSGA-II, NSGA-III, MOEA/D, SMS-EMOA, CMOPSO, and multi-objective Bayesian optimization are selected.

The selection was made to compare different approaches to multi-objective search under the same gripper-design problems and respective evaluation budgets. Including algorithms with different search principles reduces the extent to which the comparison is limited to variations of a single optimization strategy. It also allows the experiments to determine whether the relative performance depends on the selected gripper type or workpiece.

- **NSGA-II** is selected as the baseline algorithm for the population-based comparison.
- **NSGA-III** is included to compare the NSGA-II baseline with an alternative that organizes the search using predefined reference directions.
- **MOEA/D** is selected to include a decomposition-based approach and to examine how this formulation performs on the objective trade-offs of the gripper-design problems.
- **SMS-EMOA** is included as an indicator-based approach. Its selection also permits investigation of an algorithm whose search is directly associated with hypervolume, one of the performance indicators used in this thesis (Section 4.6).
- **CMOPSO** is selected to include a swarm-based search method and thereby extend the population-based comparison beyond evolutionary algorithms.
- **Multi-objective Bayesian optimization** is included as a model-based alternative to the population-based methods. It is used to assess whether a sample-efficient, surrogate-based approach is practically applicable to the investigated problems within the available computational resources.

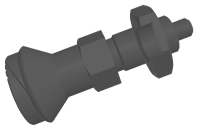
Together, these algorithms form a deliberately varied comparison set containing dominance-based, reference-direction-based, decomposition-based, indicator-based, swarm-based, and model-based approaches. The experiments use this selection to determine which approaches provide suitable convergence, trade-off representation, and consistency for the suction-cup and parallel-jaw gripper-design problems.

4.5 Workpiece Selection

Four geometrically diverse workpieces are selected to examine whether optimization behavior changes across planar, cylindrical, elongated, and free-form geometries. Such diversity is relevant because algorithm performance can depend on the feasible design space and optimization landscape [107, 108].



(a) IPA Intermediate Plate



(b) Knob



(c) Connecting rod



(d) Stanford Bunny

Figure 4.4: Selected workpieces representing planar, cylindrical, elongated, and free-form geometries.

The **IPAIntermediatePlate** (Figure 4.4(a)) is a regular industrial component with large planar regions and straight edges, representing structured, prismatic workpieces. The **Knob** (Figure 4.4(b)) is predominantly cylindrical and rotationally symmetric, with a stepped circumferential profile. The **ConnectingRod** (Figure 4.4(c)) is a flat, elongated component with a high length-to-width ratio and two circular ends connected by a narrow shaft, representing slender, asymmetric geometries. The **Stanford Bunny** (Figure 4.4(d)) provides a free-form, non-symmetric geometry with curved surfaces, irregular features, and substantial local variation.

The **Knob** is sourced from the Mechanical Components Benchmark (MCB), whereas the **Stanford Bunny** originates from the Stanford 3D Scanning Repository [109, 110]. All remaining workpieces are sourced from Fraunhofer IPA. Each workpiece defines an independent optimization instance, as described in Section 4.1.2.

4.6 Performance Indicators

Algorithm performance is evaluated using three indicators: inverted generational distance plus (IGD+), hypervolume, and total optimization runtime. Together, they characterize approximation-set quality and computational cost.

IGD+ is selected to assess how well an optimization run approximates the best trade-offs identified for a given gripper–workpiece problem. It is useful in this thesis because it considers both proximity to the reference front and the representation of different regions of that front. This enables the algorithms to be compared based on the overall quality of their approximation sets rather than on individual objective values alone.

Hypervolume is selected as an aggregate measure of approximation-set quality. It provides a single value that reflects both convergence and the extent of the represented objective-space trade-offs. Considering the final value attained during a run is particularly relevant for the investigated black-box optimization problems because it identifies the best approximation quality reached within the available optimization budget, even when the quality of the population varies during a run.

Total optimization runtime measures the computational cost of a complete run, including candidate generation, objective evaluation, and optimizer updates. It complements the quality indicators by showing the time required to obtain the resulting approximation set. Runtime comparisons are interpreted alongside the evaluation budgets, which differ between evolutionary and Bayesian optimization.

The three indicators therefore provide complementary assessments of approximation quality and computational effort. Their values across independent repetitions are used to characterize both performance and its variability.

5 Implementation

This chapter describes the software implementation used to optimize the two gripper designs. Section 5.1 describes the shared software architecture, algorithm configuration, evaluation interfaces, monitoring, and result processing. Sections 5.2 and 5.3 describe the respective geometry and grasp-evaluation implementations.

5.1 Implementation of the Common Gripper Optimization Framework

5.1.1 Software Architecture and Responsibilities

The implementation separates candidate generation from gripper evaluation. The optimizer selects a decision vector and passes it through an evaluation client to the service associated with the chosen gripper problem. The service reconstructs the design, performs the domain-specific evaluation, and returns an ordered objective vector.

The shared base class, `BaseOptimization`, provides the problem configuration, variable definitions, optimization direction, and evaluation client. Two specialized classes implement the optimization approaches:

- `EvolutionaryOptimizationProblem` connects the evaluation service to `pymoo`, using its interface for evaluating individual candidates;
- `BayesianOptimization` manages an `Ax` experiment and registers completed candidate evaluations as trials.

Here, a trial is an Ax record containing a candidate design and its evaluation results. The evaluation client is represented by the `EvaluationClient` class.

The evolutionary implementation uses a monitoring component, `HVCallback`, which records current-population hypervolume, optional cumulative-archive hypervolume (HV), and the

evaluation count after optimizer updates. The cumulative archive is optional, as its use can significantly increase computational time, particularly in optimization problems with a higher number of objectives. The Bayesian implementation uses `HypervolumeTracker`, which stores completed observations for cumulative hypervolume calculation. Both components monitor search progress without changing candidate generation or terminating a run.

Figure 5.1 summarizes the principal relationships. The evolutionary class additionally implements pymoo's `ElementwiseProblem` and the Bayesian class delegates candidate generation to the `Ax Client`. The diagram shows application-level responsibilities rather than every method or internal dependency of the external libraries.

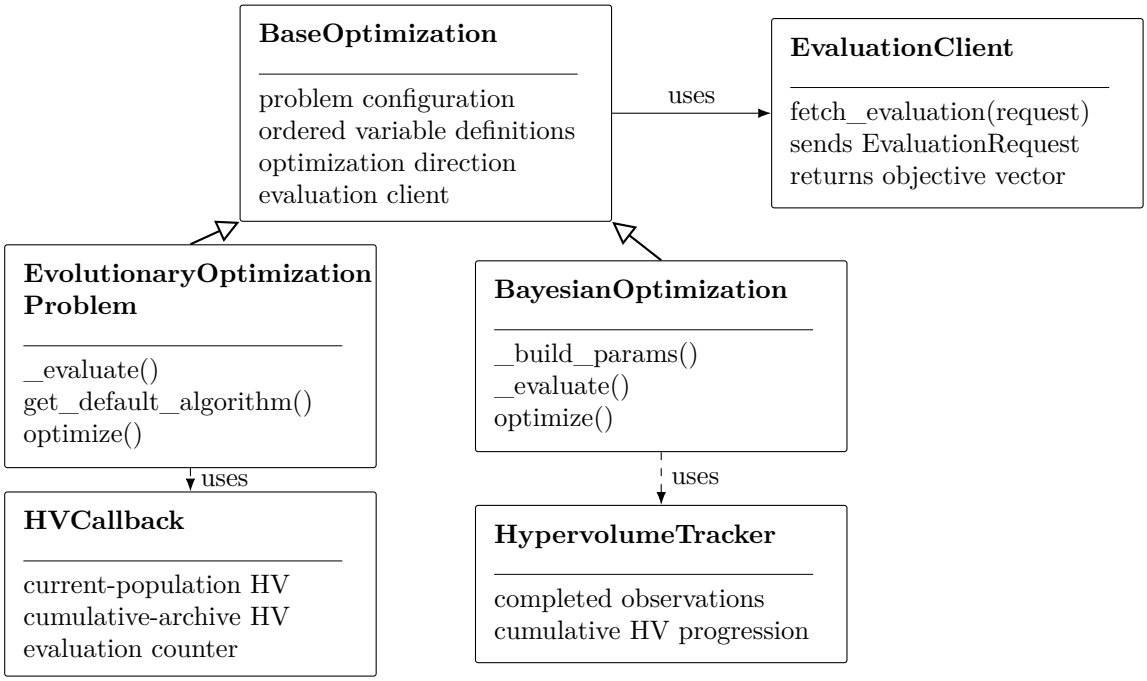


Figure 5.1: Simplified UML class diagram of the common optimization framework.

5.1.2 Configuration, Budgets, and Termination

Each optimization request specifies the workpiece, gripper problem, ordered variables and bounds, objective count, optimization method, and evaluation endpoint. Variable order is preserved when constructing the evaluation request, and returned objective values are interpreted in their configured order.

Table 5.1 summarizes the evaluation budgets and termination criteria for both optimization problems.

Evolutionary runs use a population size of 100 and stop once the configured number of objective evaluations (`n_evals`) is reached: 20,000 for the suction cup and 10,000 for the two-finger gripper. The nominal budgeting basis is population size × generations (200 and 100 generations, respectively), but the generation count is not an independent stopping criterion. Design-space tolerances, objective-space tolerances, and hypervolume (HV) stagnation are likewise not used. Consequently, the budget does not depend on how an algorithm defines a generation or how many offspring it evaluates in one update. The evaluation counts were capped separately for each gripper because computing the cumulative HV in the higher-dimensional objective space adds considerable computational overhead.

Bayesian runs request up to 10 trials per candidate-generation batch and end after 100 batches (`num_batches`), which is stored in the existing generation-count setting. The maximum budget is therefore 1,000 evaluations. Because Ax may return fewer trials than requested, and no additional trials are requested to compensate, this value is an upper bound on the number of evaluations actually completed. The limit of 100 batches was chosen because model-based candidate generation becomes increasingly time-consuming as observations accumulate, particularly beyond approximately 80 batches in the observed trial runs.

The two approaches share the evaluation pipeline but not the same total evaluation budget. In both cases, the budget is a computational-cost choice, not a convergence criterion. Progress is compared at common evaluation counts where available, and final results are interpreted together with their respective budgets and wall-clock times.

Table 5.1: Evaluation budgets and termination settings.

Problem	Algorithm	Pop./batch size	Gen./batches	Max. evals	Termination
Suction cup	Evolutionary	100	200	20,000	`n_evals`
	Bayesian	10	100	$\leq$ 1,000	`num_batches`
Two-finger	Evolutionary	100	100	10,000	`n_evals`
	Bayesian	10	100	$\leq$ 1,000	`num_batches`

5.1.3 Computing Environment

The experiments were run on four PCs (Table 5.2) under Ubuntu Linux (24.04 on PCs 1, 2 and 4; 22.04 on PC 3). The software was implemented in Python 3.12 with the packages

listed in Table 5.3. Because the GPUs differ, wall-clock times are indicative only.

Table 5.2: Hardware used for the experiments.

PC	CPU	GPU used	GPU memory	RAM
1	Intel Core i9-14900K	1× RTX 5090	32 GB	64 GB
2	Intel Core i9-10980XE	1× RTX 3060 Ti	8 GB	64 GB
3	2× Intel Xeon E5-2698 v4	2× Tesla V100	32 GB each	512 GB
4	Intel Core i9-13900K	1× RTX 3090 Ti	24 GB	64 GB

Table 5.3: Software packages and versions.

Package	Version	Package	Version
pymoo	0.6.1.6	CadQuery	2.8.0
Ax (ax-platform)	1.3.1	Open3D	0.19.0
PyTorch	2.13.0	NumPy	2.4.4
FastAPI	0.136.1	trimesh	4.12.2

5.1.4 Evolutionary Algorithm Setup

NSGA-II, NSGA-III, CMOPSO, and SMS-EMOA use a population size of 100. For MOEA/D, population size is determined by the number of reference directions rather than by a separate population-size setting. Consequently, 100 reference directions are generated to match the population size of the other algorithms. Its neighborhood size is also set to 100, covering the complete population and avoiding a narrower local restriction on information exchange.

Reference directions are required by NSGA-III and MOEA/D. Their generation method is selected according to objective count and algorithm:

- For problems with at least four objectives, the `energy` method generates 100 reference directions.
- MOEA/D always uses `energy` with 100 directions, including the three-objective suction problem.
- For NSGA-III with three objectives, `das-dennis` is used with 99 partitions.

The choice of `energy` for higher-dimensional objective spaces avoids the rapid growth of the Das–Dennis construction. For a fixed partition count, the number of Das–Dennis directions

increases rapidly with the number of objectives, substantially increasing reference-direction generation and processing costs. The `energy` method instead allows the direction count to remain fixed at 100. Both methods are briefly described in Section 2.3.9.

5.1.5 Bayesian Algorithm Setup

The Bayesian implementation creates an Ax experiment and registers the four continuous decision variables as floating-point ranges with their configured bounds.

Objectives are numbered sequentially, following the order of the vector returned by the evaluation service. All gripper objectives are configured for maximization.

The application does not specify a custom generation strategy, Sobol sample count, surrogate model, kernel, or acquisition function. Ax manages initialization and the transition to its modular BoTorch model using its defaults. These settings are therefore not presented as manually chosen experimental hyperparameters. The workflow diagram (Figure 5.3) denotes the default initialization and model-based stages as Sobol and MBM (modular BoTorch model), respectively.

For each of the 100 batches, the implementation requests up to ten new candidate trials from Ax. The returned candidates are evaluated one at a time, and each trial is immediately marked as completed with its observed objective values. The next batch is requested only after the current batch has been processed. Batch size therefore controls candidate requests, not concurrent evaluation.

5.1.6 Evaluation Interface and Objective Handling

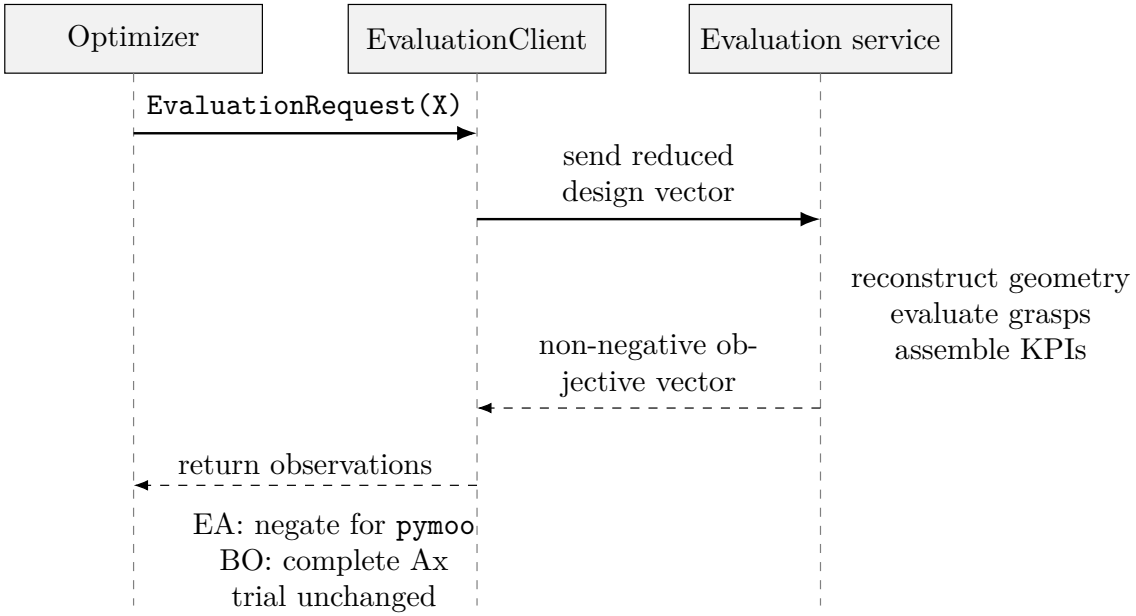


Figure 5.2: Sequence of one candidate evaluation. The optimizer waits for the returned objective vector before evaluating the next candidate. The request notation `X` denotes the ordered decision vector; EA and BO denote evolutionary and Bayesian optimization, respectively.

Figure 5.2 illustrates the evaluation of a single candidate design. The optimizer submits the ordered decision vector `X` through the `EvaluationClient` to the evaluation service, which reconstructs the geometry, evaluates the grasps, and assembles the key performance indicators (KPIs) explained in Sections 5.2.3 and 5.3.3. The resulting non-negative objective vector is returned to the optimizer. Evolutionary optimization (EA) negates these values to accommodate the minimization convention of `pymoo`, whereas Bayesian optimization (BO) reports them unchanged to Ax. The optimizer waits for the evaluation result before proceeding to the next candidate.

5.1.7 Hypervolume Monitoring

Hypervolume is calculated using `pymoo`'s indicator implementation. All indicator inputs use the minimization convention internally. The reference point has value -0.1 in every

objective dimension in the natural maximization convention. Because the original KPIs are non-negative, this reference point is worse than the evaluated outcomes.

Hypervolume Monitoring for Evolutionary optimization.

The evolutionary monitoring component records two distinct series:

- **Current-population hypervolume:** calculated based on the feasible objective vectors in the population retained at the current optimizer update. Since the hypervolume is computed exclusively from the non-dominated solutions of the current generation, without considering non-dominated solutions from previous generations, this metric may exhibit fluctuations or decreases over successive generations.
- **Cumulative-archive hypervolume:** calculated from a persistent non-dominated archive. After each optimizer update, feasible population and offspring vectors are merged into the archive, and dominated vectors are removed. This series represents the best-so-far objective coverage encountered by the monitoring component and is non-decreasing, apart from numerical round-off.

The monitoring component also records the optimizer's cumulative evaluation count. This provides the appropriate evaluation coordinate when algorithms evaluate different numbers of offspring per update.

Hypervolume Monitoring for Bayesian optimization.

For Bayesian optimization only cumulative hypervolume is calculated. After each completed trial, the tracker stores the observed objective vector. During result serialization, hypervolume is calculated for successively longer prefixes of this observation history. Each entry therefore corresponds to one completed evaluation, although the series is computed from the stored history at the end.

The tracker internally negates the maximization observations and reference point before calculating hypervolume. It does not alter the observations supplied to Ax or stored in the Bayesian frontier.

Comparisons use the same reference point and objective scaling within a given workpiece problem.

5.1.8 Result Storage and Repeated-Run Evaluation

Each run produces a JSON-compatible result containing:

- problem and optimization metadata;
- the available algorithm-configuration information;
- returned non-dominated decision vectors and objective values;
- hypervolume progression;
- total optimization wall-clock time.

The general hypervolume-history field contains the current-population series for evolutionary runs but the cumulative series for Bayesian runs. Evolutionary results additionally contain a separate cumulative-archive hypervolume field. The stored series is therefore interpreted according to the optimizer family.

The evolutionary frontier consists of the decision vectors and corresponding objective values returned by the optimizer at termination. It must not be confused with the monitoring component's cumulative archive: the final returned frontier can have lower hypervolume than the best-so-far archive. The complete frontier archive is not stored due to computational limitations. For Bayesian optimization, Ax identifies the Pareto-frontier trials. The implementation uses the stored observations associated with those trials where available. Otherwise, it uses the objective metric means returned by Ax.

These outputs are finite-search approximations rather than guarantees of the global Pareto front. Since the true reference front is unknown, IGD+ is evaluated against an empirical non-dominated reference set assembled from repeated runs of the same workpiece. This provides a within-problem measure of the coverage of the reference approximation, with lower values preferred. It is not an absolute distance to the unknown true Pareto front, and values based on different empirical reference sets are not directly comparable.

5.1.9 Summary of the Optimization Workflow

Figure 5.3 illustrates the general optimization workflow described in this section, encompassing configuration, evaluation, monitoring, termination, and result storage. Both optimization branches rely on the same problem-specific evaluation service but operate with different

evaluation budgets. Dashed arrows indicate requests sent to and responses received from the shared evaluation API.

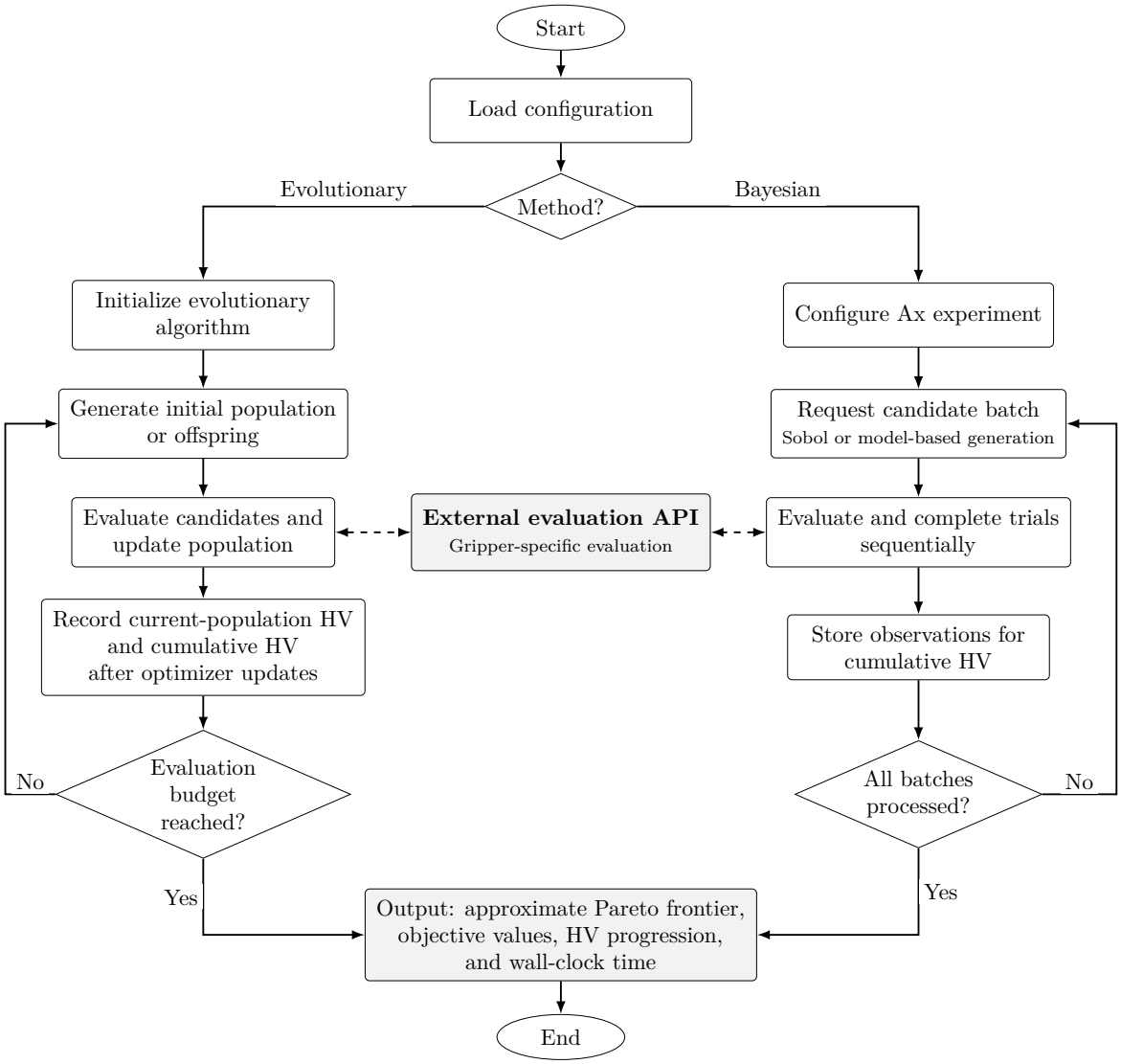


Figure 5.3: Overview of the evolutionary (`pymoo`) and Bayesian (`Ax`/`BoTorch`) optimization workflows.

5.2 Implementation of Multi-Suction Cup Gripper Optimization

5.2.1 Decision Variables and Bounds for Multi-Suction Cup Gripper

The optimization is performed over the reduced decision vector and all four variables are continuous.

The decision variables and their configured bounds are summarized in Table 5.4. For a suction-cup diameter of 0.01 m (adopted from the suction-cup configuration employed in the bin-picking demonstration conducted by Fraunhofer IPA), the configured lower radial bound is 0.005 m for all workpieces and is calculated using Eq. 5.3. The upper radial bound depends on the workpiece. All angular variables use the interval $[0,2\pi)$.

Table 5.4: Decision variables and configured bounds for multi-suction-cup gripper optimization with a suction-cup diameter of 0.01 m.

Variable	**Workpiece**	**Unit**	**Lower bound**	**Upper bound**
r	Bunny	m	0.005	0.048
r	IPAIntermediatePlate	m	0.005	0.041
r	IPAConnectingRod	m	0.005	0.033
r	MCBA_Knob	m	0.005	0.031
α_1	All workpieces	rad	0	2π
α_2	All workpieces	rad	0	2π
α_3	All workpieces	rad	0	2π

The radial search interval is expressed as

$$r \in [r_{\min}, r_{\max}]. \tag{5.1}$$

The radial bounds are determined separately for each workpiece. The workpiece is oriented such that its longest dimension lies in the horizontal plane, the horizontal bounding box is centered at $(x,y) = (0,0)$, and the base of the workpiece is placed at $z = 0$. This establishes a consistent coordinate system for defining the gripper-design bounds.

$$\begin{aligned} r_{\max} &= \min\left\{|X_{\min}|, |X_{\max}|, |Y_{\min}|, |Y_{\max}|\right\}, \\ X_{\min}, X_{\max}, Y_{\min}, Y_{\max} &\in \mathbb{R}, \\ X_{\min} \le X_{\max}, &\quad Y_{\min} \le Y_{\max}, \\ r_{\max} &\in \{r \in \mathbb{R} \mid r \ge 0\}. \end{aligned} \tag{5.2}$$

The geometric definition of the lower radial bound is based on the suction-cup radius r_{cup}:

$$r_{\min} = \frac{2\sqrt{3}}{3}\, r_{\mathrm{cup}}, \qquad r_{\mathrm{cup}}, r_{\min} \in \{r \in \mathbb{R} \mid r \geq 0\}\,. \tag{5.3}$$

This lower bound corresponds to the circumradius of an equilateral triangle with side length $2r_{\mathrm{cup}}$, i.e. the configuration in which neighboring cups just touch.

All optimization methods generate candidate values within these bounds. In the evolutionary algorithms, the initial population is obtained by random sampling and subsequently modified through the operators of the selected algorithm. In Bayesian optimization, Ax first evaluates the center of the search space and then uses Sobol sampling to generate an initial space-filling design. After sufficient initialization data have been collected, Ax transitions to model-based Bayesian optimization.

5.2.2 Evaluation of Multi-Suction Cup Gripper

Each candidate is sent to the external evaluation API as the reduced decision vector $\mathbf{q}_{\mathrm{SC}} = [r,\alpha_1,\alpha_2,\alpha_3]$. The server maps this vector to the Cartesian cup configuration $\mathbf{P} = \Phi(\mathbf{q}_{\mathrm{SC}})$, generates and filters grasps, and returns the objectives defined in 5.2.3.

Algorithm 1 summarizes one evaluation request.

Algorithm 1 API evaluation of a multi-suction cup gripper candidate

Require: Candidate $\mathbf{q}_{\mathrm{SC}}$, fixed coordinate z_0, and configured workpiece
Ensure: Objective vector $\mathbf{f}(\mathbf{q}_{\mathrm{SC}})$
1: Send $\mathbf{q}_{\mathrm{SC}}$ to the evaluation API
2: **for** $i = 1,2,3$ **do**
3: Construct cup position $\mathbf{p}_i = (r\cos(\alpha_i), r\sin(\alpha_i), z_0)$
4: **end for**
5: Assemble the Cartesian configuration $\mathbf{P} = \Phi(\mathbf{q}_{\mathrm{SC}})$ from the cup positions
6: Generate suction grasp candidates for $\mathbf{P}$
7: Retain grasps that pass seal-formation and collision checks
8: Cluster the retained grasps
9: Compute D, $\hat{n}_{\mathrm{g}}$, and $\hat{A}$
10: Return $\mathbf{f}(\mathbf{q}_{\mathrm{SC}}) = [D,\hat{n}_{\mathrm{g}},\hat{A}]$ to the selected optimization framework

Grasp Evaluation

The evaluation service constructs the Cartesian gripper configuration $\mathbf{P} = [\mathbf{p}_1,\mathbf{p}_2,\mathbf{p}_3]$ from the received decision vector and computes the initial grasp candidates, performs collision check using a gripper collider model and clusters the valid grasps through the following procedure:

Candidate Grasp Generation and Feasibility Filtering

Suction grasp candidates are generated using a method developed by Fraunhofer IPA, following the procedure described in [84]. A total of 1,000 points are uniformly sampled across the surface of the workpiece to obtain a sufficiently dense spatial distribution of potential grasp locations. Each sampled point defines a candidate grasp characterized by its position and the corresponding local surface normal. The surface normal determines the approach direction of the suction cup.

In the original method, each suction grasp candidate is tested for feasible seal formation using a quasi-static spring model of a suction cup, as introduced in [85]. If the grasp point is considered feasible, the candidate is further tested with a gripper collider model for collisions between the gripper and the object.

In this thesis, the method is extended to a gripper with three suction cups. For each candidate gripper pose, the seal-formation test is performed for all three suction cup contact positions $\mathbf{p}_1$, $\mathbf{p}_2$, and $\mathbf{p}_3$. A candidate is considered feasible only if all three suction cups form a sufficient seal. Each feasible candidate is subsequently subjected to a geometric collision check: the gripper geometry is represented by a sampled point set, transformed into the workpiece frame by the candidate pose, and tested for proximity against the workpiece point cloud. A candidate is rejected if any transformed gripper point lies closer than the clearance threshold of $\delta_{\text{clr}} = 0.5\,\text{mm}$ (adopted from the bin-picking demonstration conducted by Fraunhofer IPA) to any workpiece point. The candidates that pass both the seal-formation and the collision checks form the set of valid grasps $\mathcal{H}_{\text{valid}}$, which is the input to the subsequent clustering step.

Clustering of Valid Grasps

The set $\mathcal{H}_{\text{valid}}$ typically contains many grasps that are nearly identical in position and orientation. If the number of valid grasps $n_{\text{v}} = |\mathcal{H}_{\text{valid}}|$ exceeds the clustering budget $K_{\text{clst}} = 15$,

the grasps are clustered using the greedy farthest-point sampling (FPS) algorithm of [111], which iteratively selects the grasp with the largest minimum distance to the already selected representatives. Otherwise, all valid grasps are retained. The value of $K_{\text{clst}} = 15$ is selected to strike a balance between computational cost and obtaining a sufficiently large number of candidate grasps for meaningful diversity calculations.

Since grasp poses combine a translational and a rotational component, the sampling operates on a combined pose metric in which the Euclidean distance between the grasp positions $\mathbf{x}_i \in \mathbb{R}^3$ is normalized by the diagonal extent of the axis-aligned bounding box of all candidates, and the difference between the orientations $\mathbf{R}_i \in SO(3)$ is measured as the geodesic angle normalised by π. With the resulting normalized terms $\hat{\rho}_{\text{pos}}$ and $\hat{\rho}_{\text{ori}}$ and an orientation weight $w \in [0,1]$,

$$\begin{gathered} \rho_w(i,j) = (1-w)\,\hat{\rho}_{\text{pos}}(i,j) + w\,\hat{\rho}_{\text{ori}}(i,j), \\ w \in [0,1], \\ \hat{\rho}_{\text{pos}}(i,j), \hat{\rho}_{\text{ori}}(i,j), \rho_w(i,j) \in [0,1], \quad \forall\, i,j \in \{1,\ldots,N\}, \end{gathered} \tag{5.4}$$

so that $w = 0$ corresponds to purely position-based and $w = 1$ to purely orientation-based sampling. Here $w = 0.35$ is chosen to balance positional and orientational differences, with a greater emphasis on positional variation. All distances are evaluated on the GPU, which keeps the selection tractable for large candidate sets.

The final number of representatives K^* is not predefined but obtained from an elbow criterion on the covering radius recorded during sampling: selection stops as soon as the relative coverage gain of an additional representative falls below $\tau = 0.05$, bounded from above by K_{clst}. The resulting subset $\mathcal{H}_{\text{clst}} \subseteq \mathcal{H}_{\text{valid}}$ with $|\mathcal{H}_{\text{clst}}| = K^*$ is passed on to the diversity evaluation.

5.2.3 Objective Evaluation of Multi-Suction Cup Gripper

After grasp evaluation, each candidate layout is assigned the natural objective vector

$$\mathbf{f}(\mathbf{q}_{\text{SC}}) = \left[D(\mathbf{P}),\, \hat{n}_{\text{g}}(\mathbf{P}),\, \hat{A}(\mathbf{P}) \right], \qquad \mathbf{P} = \Phi(\mathbf{q}_{\text{SC}}). \tag{5.5}$$

The three objectives are calculated as follows:

Diversity Metric Calculation ($D(\mathbf{P})$)

The diversity score combines spatial and rotational diversity:

$$D = \begin{cases} 0, & n_\mathrm{g} < 2, \\ \mathrm{clip}_{[0,1]}(\lambda D_\mathrm{pos} + (1-\lambda) D_\mathrm{rot}), & n_\mathrm{g} \geq 2, \end{cases}$$

where

$$\begin{aligned} D_\mathrm{pos} &= (1-\beta_\mathrm{cov}) D_\mathrm{spread} + \beta_\mathrm{cov} D_\mathrm{cov}, \\ D_\mathrm{rot} &= \frac{1}{\pi M} \sum_{(i,j)\in\mathcal{I}} \rho_\mathrm{geo}(\mathbf{R}_i, \mathbf{R}_j), \end{aligned}$$

where

$$\begin{aligned} D_\mathrm{spread} &= \frac{1}{Ms} \sum_{(i,j)\in\mathcal{I}} \|\mathbf{y}_i - \mathbf{y}_j\|_2, \\ D_\mathrm{cov} &= \mathrm{clip}_{[0,1]}\left(\frac{1}{2M} \sum_{(i,j)\in\mathcal{I}} \left(1 - \hat{\mathbf{u}}_i^\mathsf{T} \hat{\mathbf{u}}_j\right) \right), \\ n_\mathrm{g} &\in \{n \in \mathbb{Z} \mid n \geq 0\}, \\ \lambda, \beta_\mathrm{cov} &\in \{r \in \mathbb{R} \mid 0 \leq r \leq 1\}, \\ D, D_\mathrm{pos}, D_\mathrm{spread}, D_\mathrm{cov}, D_\mathrm{rot} &\in \{r \in \mathbb{R} \mid 0 \leq r \leq 1\}. \end{aligned} \tag{5.6}$$

For $n_\mathrm{g} < 2$, the diversity score is set to zero, because at least two grasps are required for a pairwise comparison. For $n_\mathrm{g} \geq 2$, all pairwise terms are averaged over the set of unique grasp pairs $\mathcal{I}$, which contains M elements:

$$\begin{aligned} \mathcal{I} &= \left\{ (i,j) \in \mathbb{N}^2 \,\middle|\, 1 \leq i < j \leq n_\mathrm{g} \right\}, \\ M = |\mathcal{I}| &= \frac{n_\mathrm{g}(n_\mathrm{g}-1)}{2}, \\ M &\in \{m \in \mathbb{Z} \mid m \geq 1\}. \end{aligned} \tag{5.7}$$

The spatial terms use the grasp points $\mathbf{y}_i$ projected onto the workpiece surface $\partial\mathcal{W}$. D_spread normalizes their mean pairwise distance by the bounding-box diagonal s, where $\mathbf{e}$ contains the bounding-box side lengths. D_cov uses the direction vectors $\hat{\mathbf{u}}_i$ from the workpiece centroid $\mathbf{o}$:

$$
\begin{aligned}
\mathbf{y}_i &= \arg\min_{\boldsymbol{\xi}\in\partial\mathcal{W}} \|\boldsymbol{\xi} - \mathbf{x}_i\|_2, \\
s &= \|\mathbf{e}\|_2, \\
\hat{\mathbf{u}}_i &= \frac{\mathbf{y}_i - \mathbf{o}}{\|\mathbf{y}_i - \mathbf{o}\|_2 + \varepsilon}, \\
\mathbf{x}_i, \mathbf{y}_i, \hat{\mathbf{u}}_i, \mathbf{o} &\in \mathbb{R}^3, \quad \partial\mathcal{W} \subset \mathbb{R}^3, \\
\mathbf{y}_i &\neq \mathbf{o}, \quad \forall\, i \in \{1,\ldots,n_{\mathrm{g}}\}, \\
\mathbf{e} &\in \left\{ \mathbf{v} \in \mathbb{R}^3 \mid v_k > 0,\, k = 1,2,3 \right\}, \\
s, \varepsilon &\in \{ r \in \mathbb{R} \mid r > 0 \}.
\end{aligned}
\tag{5.8}
$$

The spread term rewards distance between grasps. The coverage term separates grasps distributed around the object from grasps concentrated in one region.

The rotational diversity D_{rot} is the mean pairwise geodesic distance between the grasp orientations, normalized by its maximum value π:

$$
\begin{aligned}
\rho_{\mathrm{geo}}(\mathbf{R}_i,\mathbf{R}_j) &= \cos^{-1}\Big(\mathrm{clip}_{[-1,1]}(q_{ij})\Big), \\
q_{ij} &= \frac{\mathrm{tr}(\mathbf{R}_i^{\mathsf{T}}\mathbf{R}_j) - 1}{2}, \\
\mathbf{R}_i, \mathbf{R}_j &\in SO(3), \quad q_{ij} \in \mathbb{R}, \\
\rho_{\mathrm{geo}}(\mathbf{R}_i,\mathbf{R}_j) &\in \{ r \in \mathbb{R} \mid 0 \le r \le \pi \}.
\end{aligned}
\tag{5.9}
$$

The clipping function keeps the inverse-cosine argument in its valid range, which guards against floating-point errors:

$$
\begin{aligned}
\mathrm{clip}_{[a,b]} &: \mathbb{R} \to \{ r \in \mathbb{R} \mid a \le r \le b \}, \\
z &\mapsto \min(b, \max(a,z)), \\
a, b, z &\in \mathbb{R}, \quad a \le b.
\end{aligned}
\tag{5.10}
$$

Normalized Grasp Count ($\hat{n}_{\mathrm{g}}(\mathbf{P})$)

The grasp-count objective measures the number of grasps retained for evaluation, rather than the total number of feasible candidates. It rewards a larger retained grasp set under

the configured clustering procedure. The count is taken from the same set used for diversity evaluation:

$$n_{\mathrm{g}} = \begin{cases} |\mathcal{H}_{\mathrm{clst}}|, & \text{if clustering is performed,} \\ |\mathcal{H}_{\mathrm{valid}}|, & \text{otherwise.} \end{cases} \tag{5.11}$$

To keep this quantity comparable with the remaining objectives, it is expressed relative to the clustering budget K_{clst}, which is the maximum number of grasps retained by the clustering stage,

$$\begin{aligned} \hat{n}_{\mathrm{g}} &= \frac{n_{\mathrm{g}}}{K_{\mathrm{clst}}}, \\ K_{\mathrm{clst}} &\in \{k \in \mathbb{Z} \mid k \geq 1\}, \\ n_{\mathrm{g}} &\in \{n \in \mathbb{Z} \mid 0 \leq n \leq K_{\mathrm{clst}}\}, \\ \hat{n}_{\mathrm{g}} &\in \{r \in \mathbb{R} \mid r \in [0,1]\}. \end{aligned} \tag{5.12}$$

Since clustering is only applied when more than K_{clst} valid grasps exist, the condition $n_{\mathrm{g}} \leq K_{\mathrm{clst}}$ holds in both cases and the ratio is bounded by unity without further clipping. A value of one indicates that the layout produced at least as many valid grasps as the clustering budget permits, whereas a value of zero corresponds to a layout for which no valid grasp was found.

Normalized Area Calculation ($\hat{A}(\mathbf{P})$)

The area of the triangle formed by the three suction cups is computed from their Cartesian coordinates. With the planar coordinates $\mathbf{P} = (p_{i,x}, p_{i,y})$ of cup i,

$$\begin{aligned} A(\mathbf{P}) &= \frac{1}{2}\big|(p_{2,x} - p_{1,x})(p_{3,y} - p_{1,y}) \\ &\qquad - (p_{3,x} - p_{1,x})(p_{2,y} - p_{1,y})\big|, \\ A_{\max} &= \tfrac{3\sqrt{3}}{4}\, r_{\max}^2, \\ A : \mathcal{P}_{\mathcal{W}} &\to \{r \in \mathbb{R} \mid 0 \leq r \leq A_{\max}\}, \quad \mathbf{P} \in \mathcal{P}_{\mathcal{W}}, \\ \mathbf{p}_k = (p_{k,x}, p_{k,y})^{\mathsf{T}} &\in \left\{\mathbf{v} \in \mathbb{R}^2 \mid \|\mathbf{v}\|_2 \leq r_{\max}\right\}, \quad \forall\, k \in \{1,2,3\}, \\ r_{\max} &\in \{r \in \mathbb{R} \mid r \geq 0\}. \end{aligned} \tag{5.13}$$

The area is zero exactly when the three cups lie on one line. It reaches $A_{\max}$ when they form an equilateral triangle on a circle of radius $r_{\max}$, where $r_{\max}$ in the expression for $A_{\max}$ is set to the value of the Stanford Bunny, 0.048 m

To make this objective comparable in magnitude with the remaining objectives and across workpieces, the area is normalized by a single global constant. $A_{\max}$ is calculated for the largest $r_{\max}$ among all four workpieces. The maximum-area triangle inscribed in a circle is the equilateral one,

$$\begin{aligned} A_{\max} &= \frac{3\sqrt{3}}{4}\, r_{\max}^2, \\ r_{\max} &= r_{\max}^{\text{Bunny}} = 0.048\,\text{m}, \\ A_{\max} &\in \{r \in \mathbb{R} \mid r \geq 0\}\,. \end{aligned} \tag{5.14}$$

Here, $r_{\max}$ is set to the value of the Stanford Bunny, giving $A_{\max} \approx 2.993 \times 10^{-3}\,\text{m}^2$. This value is used for all workpieces. Elsewhere in this thesis, $r_{\max}$ denotes the workpiece-specific upper radial bound.

The normalized area objective is therefore

$$\hat{A} = \frac{A}{A_{\max}} \in [0,1]. \tag{5.15}$$

This objective favors gripper layouts with a wider support polygon, which is associated with the assumption of improved grasp stability. Because the cups are restricted to the $r_{\max}$ of the respective workpiece, $\hat{A}$ can reach 1 only for the Stanford Bunny. For the other workpieces, the largest attainable value is $(r_{\max}/0.048\,\text{m})^2$, using the workpiece-specific $r_{\max}$: approximately 0.730 (IPAIntermediatePlate), 0.473 (IPAConnectingRod), and 0.417 (MCBA_Knob).

Continuity considerations : The suction objectives are not guaranteed to be continuous in the decision variables. Small layout changes can alter seal feasibility, collision classification, the retained grasp count, or the representatives selected by clustering. Abrupt changes can also occur when the clustering threshold or elbow stopping criterion is crossed.

If fewer than two grasps remain for diversity evaluation, the pairwise diversity components are set to zero. Thus, the evaluator samples 1,000 initial surface points to provide a dense candidate set. This reduces dependence on sparse sampling but does not guarantee continuous objectives or the existence of valid grasps.

5.3 Implementation of Two-Finger Gripper Optimization

This section describes the problem-specific implementation for a two-finger gripper with a spline-based gripping contour. The common framework components are described in Section 5.1.

5.3.1 Decision Variables and Bounds for Two-Finger Gripper Problem

The decision variables and their bounds for two-finger gripper are summarized in Table 5.5.

Table 5.5: Decision variables of the two-finger gripper optimization problem.

Variable	Lower bound (mm)	Upper bound (mm)
ℓ_{gn}	20	50
h_1	2	10
h_2	2	10
h_3	2	10

All four decision variables are treated as continuous. The bounds were selected to ensure that the contour length remains sufficiently large relative to the depth offsets, while avoiding physically infeasible jaw contours.

For evolutionary optimization, the variables are initially sampled within these bounds and subsequently modified by the operators of the selected algorithm. For Bayesian optimization, Ax initially evaluates the center of the search space and generates additional initialization candidates using Sobol sampling. After sufficient observations have been collected, Ax transitions to model-based candidate generation.

5.3.2 Evaluation of Two-Finger Gripper

Each candidate decision vector $\mathbf{q}_{\mathrm{PJ}}$ is sent as an ordered list to the evaluation API. The mapping $\mathbf{c}_{\mathrm{gr}}(\cdot)$ inserts the decision variables into a complete gripper configuration, and $\mathcal{G}(\mathbf{q}_{\mathrm{PJ}})$ denotes the resulting jaw geometry.

Algorithm 2 summarizes one evaluation request.

Algorithm 2 API evaluation of a two-finger gripper candidate

Require: Candidate $\mathbf{q}_{\mathrm{PJ}}$, fixed gripper parameters, and configured workpiece with predefined orientations
Ensure: One proximity KPI per predefined workpiece orientation
1: Send $\mathbf{q}_{\mathrm{PJ}}$ to the evaluation API as an ordered list
2: Map $\mathbf{q}_{\mathrm{PJ}}$ to the complete configuration $\mathbf{c}_{\mathrm{gr}}(\mathbf{q}_{\mathrm{PJ}})$
3: Generate the spline-based jaw geometry $\mathcal{G}(\mathbf{q}_{\mathrm{PJ}})$ using CadQuery
4: **for all** predefined workpiece orientations **do**
5: Voxelize the gripper and the workpiece in the current orientation
6: Calculate FFT-based collision maps using the modified Snuggle-Pack procedure
7: Calculate and store the proximity KPI for the current orientation
8: **end for**
9: Return the KPI vector in predefined orientation order to the selected optimization framework

Grasp Set Definition

A grasp specifies the pose in which the workpiece is presented to the jaws. In the evaluation pipeline, the gripper remains stationary in its nominal pose, and the workpiece is reoriented for each grasp.

Only the orientation component of the generated grasp poses is used, since the relative gripper–workpiece position is determined by the collision- and proximity-based placement search described in Section 5.3.2. A grasp γ_i is therefore fully described by a spatial rotation $\mathbf{R}_i$ applied to the workpiece,

$$\begin{gathered} \mathcal{W}_i = \mathbf{R}_i\,\mathcal{W}, \qquad i = 1,\ldots,n_{\mathrm{or}}, \\ \mathbf{R}_i \in \mathrm{SO}(3), \quad \mathcal{W}_i \subset \mathbb{R}^3, \quad \forall\, i \in \{1,\ldots,n_{\mathrm{or}}\}, \\ \mathcal{W} \subset \mathbb{R}^3, \quad n_{\mathrm{or}} \in \{n \in \mathbb{Z} \mid n \geq 1\}\,. \end{gathered} \tag{5.16}$$

where $\mathcal{W}$ denotes the workpiece in its reference orientation. The translation components of the generated poses are discarded. The grasp set is

$$\mathcal{R} = \{\mathbf{R}_1,\ldots,\mathbf{R}_{n_{\mathrm{or}}}\}\,, \qquad n_{\mathrm{or}} = 4. \tag{5.17}$$

In the present study, $n_{\mathrm{or}} = 4$ orientations are configured. This value is selected as a compromise between the dimensionality of the optimization problem and retaining sufficient directional diversity to evaluate the grasping performance across the workpiece geometry.

The predefined grasps are a spatially distributed subset of the physically admissible grasps of the given workpiece, which ensures that the evaluated orientations are mutually dissimilar and at the same time realizable on the part geometry.

The grasp set is generated once before the optimization starts and is stored as part of the workpiece list. It is therefore identical for all candidates and for all repetitions of a run, which makes the objective values of different candidates directly comparable. The orientations are fixed quantities and are not optimization variables.

Grasp Evaluation

Following the proximity criterion in Section 4.2.3, the gripper is evaluated for each grasp γ_i, $i = 1,\ldots,n_{\mathrm{or}}$. The gripper remains in its assembly pose while the workpiece is rotated to the corresponding orientation. Both geometries are voxelized and gripper surface voxels are closed by morphological dilation and hole filling, with part labels retained to distinguish the jaws.

The modified Snuggle-Pack procedure computes an overlap map for each jaw $j \in \{\mathrm{L},\mathrm{R}\}$ from its occupancy grid $\chi_{\mathcal{J}_j}$ and the oriented workpiece grid $\chi_{\mathcal{W}_i}$:

$$\begin{gathered}\zeta_{i,j} = \mathrm{DFT}^{-1}\Big(\mathrm{DFT}(\chi_{\mathcal{J}_j})\ \mathrm{DFT}(\chi_{\mathcal{W}_i})\Big), \qquad \zeta_{i,j}(\mathbf{v}) < \varepsilon_{\mathrm{col}} \iff \mathbf{v} \text{ is collision-free,} \\ \forall\, i \in \{1,\ldots,n_{\mathrm{or}}\}, \quad \forall\, j \in \{\mathrm{L},\mathrm{R}\}, \quad \forall\, \mathbf{v} \in \mathbb{Z}^3, \\ \chi_{\mathcal{J}_j}, \chi_{\mathcal{W}_i} : \mathbb{Z}^3 \to \{0,1\}, \quad \zeta_{i,j} : \mathbb{Z}^3 \to \mathbb{R}, \\ \varepsilon_{\mathrm{col}} \in \{r \in \mathbb{R} \mid 0 < r < 1\}\,.\end{gathered} \tag{5.18}$$

Here, $\mathbf{v}$ is a discrete displacement and $\varepsilon_{\mathrm{col}}$ absorbs single-precision FFT round-off. Collision freedom refers only to the voxelized jaw model. The gripper base is excluded. Padding to a common grid size makes the transform cyclic.

Each overlap map is convolved with a one-sided Gaussian kernel along the jaw's closing direction, favoring placements where a small closing motion would establish contact. Kernel standard deviations are converted from millimeters to voxel units using the voxel edge length. Only the width and height deviations affect the score because the kernel acts in the plane orthogonal to the gripper length axis. Colliding displacements are excluded.

For each position in the length–height plane, the jaws independently select their best collision-free closing-axis positions. Maximizing their summed proximity over the plane

yields p_i: both jaws share the length–height placement but move independently along the closing axis. FFT computations use GPU-accelerated PyTorch. The score is dimensionless, unnormalized, and Gaussian-weighted. It measures neither distance nor voxel count and depends on workpiece geometry and voxel resolution.

Figure 5.4 shows the combined proximity map for one workpiece orientation. For each length–height grid index, the implementation selects the highest collision-free proximity value along the closing axis independently for each jaw and then adds the two values. Brighter finite values represent higher combined scores. The maximum of the map gives p_i before the opening-width feasibility test is applied.

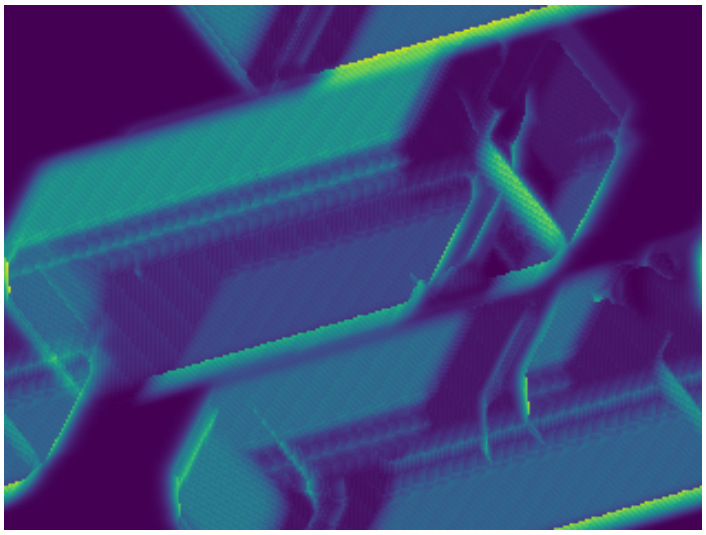


Figure 5.4: Combined jaw proximity map.

The difference between the selected jaw positions defines the signed opening width δ_i. Negative values indicate closing onto the workpiece, while non-negative values require opening beyond the modeled position. Only the sign is tested, not the magnitude relative to the stroke. The returned KPI is

$$
\begin{aligned}
k_i(\mathbf{q}_{\mathrm{PJ}}) &= \begin{cases} p_i(\mathbf{q}_{\mathrm{PJ}})/100, & \delta_i(\mathbf{q}_{\mathrm{PJ}}) < 0, \\ 0, & \delta_i(\mathbf{q}_{\mathrm{PJ}}) \geq 0, \end{cases} \qquad i = 1, \ldots, n_{\mathrm{or}}, \\
p_i(\mathbf{q}_{\mathrm{PJ}}) &\in \{r \in \mathbb{R} \mid r \geq 0\}, \quad \forall\, i \in \{1, \ldots, n_{\mathrm{or}}\}, \\
\delta_i(\mathbf{q}_{\mathrm{PJ}}) &\in \mathbb{R}, \quad k_i(\mathbf{q}_{\mathrm{PJ}}) \in \{r \in \mathbb{R} \mid r \geq 0\}, \quad \forall\, i \in \{1, \ldots, n_{\mathrm{or}}\}.
\end{aligned}
\tag{5.19}
$$

Initial unscaled KPI values typically ranged from 0 to 200. Division by 100 reduces objective

magnitudes and makes the resulting hypervolume numerically well-scaled.

Feasibility is assessed independently per grasp. An infeasible orientation receives zero, the worst non-negative objective value, rather than imposing a hard constraint on the entire candidate.

5.3.3 Objective Evaluation of Two-Finger Gripper

The evaluation service returns one grasp-quality KPI for each configured grasp. Since each optimization run considers a single workpiece and its associated set of grasps, the number of objectives is directly determined by the size of the grasp set. The individual KPIs are not aggregated into a scalar value and each grasp retains its own objective, which preserves the information about which orientations a given contour favors.

Since a larger proximity KPI represents a better grasp result, the natural optimization problem is

$$\begin{aligned} \max_{\mathbf{q}_{\mathrm{PJ}} \in \mathcal{Q}} k_i(\mathbf{q}_{\mathrm{PJ}}), \qquad & i = 1, \ldots, n_{\mathrm{or}}, \\ \mathcal{Q} \subset \mathbb{R}^4, \qquad & n_{\mathrm{or}} = 4, \\ k_i : \mathcal{Q} \to \mathcal{K} \subseteq \mathbb{R}_{\geq 0} & \end{aligned} \tag{5.20}$$

with the objective vector

$$\mathbf{f}(\mathbf{q}_{\mathrm{PJ}}) = \begin{bmatrix} k_1(\mathbf{q}_{\mathrm{PJ}}) & k_2(\mathbf{q}_{\mathrm{PJ}}) & \cdots & k_{n_{\mathrm{or}}}(\mathbf{q}_{\mathrm{PJ}}) \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{n_{\mathrm{or}}}_{\geq 0}. \tag{5.21}$$

Because the four decision variables define a single jaw contour that is shared by all grasps, the objectives are generally conflicting: a contour whose depth profile conforms closely to the workpiece in one orientation does not necessarily conform in the remaining ones. The optimization therefore yields a Pareto-front approximation that represent different compromises in geometric conformity across the predefined orientations.

6 Results

This chapter reports the evolutionary optimization results for the suction-cup (SC) and two-finger grippers. Bayesian optimization yielded some results, but these were limited in their usefulness. Each gripper–workpiece pair is an independent optimization instance (Section 4.1.2), so results are reported per workpiece and are not pooled across workpieces or grippers.

The indicators are used as follows:

- **Hypervolume (HV)** is compared across algorithms within a workpiece, because all runs share the same reference point.
- **IGD+** is computed against one shared reference front per workpiece. This front is built from the final solutions of all algorithms and runs on that workpiece. IGD+ is compared across algorithms within a workpiece, but not across workpieces.
- **Runtime** was measured on GPUs with different computing capacity. It is reported for information only and is not used for ranking.

All values are reported as mean ± sample standard deviation over independent runs. In the tables, the best mean for IGD+ and HV is shown in bold.

6.1 Suction-Cup Gripper Results

6.1.1 Preliminary Two-Objective Runs

Initial test runs used two objectives, the normalized triangle area $\hat{A}$ and the grasp diversity D. The resulting Pareto solutions contained only two grasps, mostly located at opposite ends of the workpiece (Figure 6.1). Two grasps at opposite ends already give a high diversity score, so maximizing D favored layouts with exactly two grasps and gave no incentive to retain additional distinct grasps. For this reason, the normalized grasp count $\hat{n}_g$ was added

as a third objective (Section 4.2.2). All suction-cup results below use this three-objective formulation.

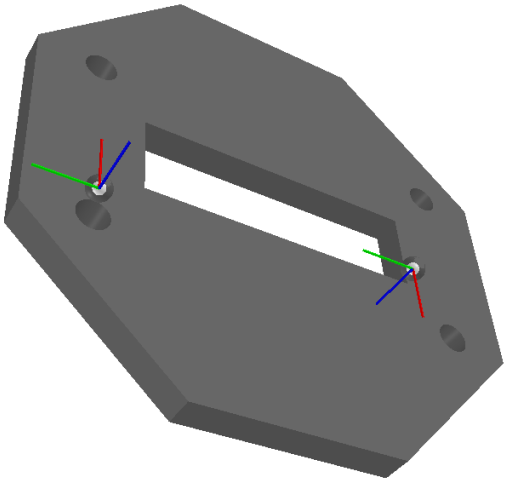


Figure 6.1: Grasp points on the IPA Intermediate Plate from the two-objective suction-cup run (area and diversity).

6.1.2 Experimental Settings

Each evolutionary algorithm was run five times per workpiece, with a population of 100, 200 generations, and a budget of 20,000 evaluations (Table 5.1). The archive HV is the best-so-far cumulative archive at termination. The reference point is -0.1 in the natural maximization convention, which corresponds to 0.1 in the minimization convention used internally by the indicator.

6.1.3 Individual Results per Workpiece

IPA Intermediate Plate

NSGA-II and SMS-EMOA form the leading group on this workpiece. SMS-EMOA gives the lowest IGD+, and NSGA-II the highest archive HV, with SMS-EMOA nearly tied in HV. SMS-EMOA is the only algorithm that combines the lowest IGD+ with an HV close to the best. CMOPSO and NSGA-III form a middle group in HV, and MOEA/D is the weakest on both indicators. The leading algorithms show a small run-to-run spread, whereas the IGD+ of MOEA/D varies more between runs (Table 6.1).

Table 6.1: Suction-cup results: IPA Intermediate Plate.

Algorithm	**IGD+ ↓**	**Archive HV ↑**	**Runtime (s)**
CMOPSO	0.0131 ± 0.0007	0.3626 ± 0.0045	102500.4 ± 2099.4
MOEA/D	0.0312 ± 0.0053	0.3515 ± 0.0084	104270.6 ± 2469.3
NSGA-II	0.0122 ± 0.0018	$\mathbf{0.3668 \pm 0.0011}$	71979.8 ± 785.4
NSGA-III	0.0133 ± 0.0012	0.3604 ± 0.0057	144361.2 ± 1115.4
SMS-EMOA	$\mathbf{0.0067 \pm 0.0003}$	0.3664 ± 0.0012	67575.8 ± 751.0

Knob

NSGA-III, NSGA-II, and SMS-EMOA form a leading group that is close on both indicators. NSGA-III has the highest archive HV, while SMS-EMOA has the lowest IGD+ of the three, so no single algorithm leads on both. CMOPSO is clearly behind this group in HV. MOEA/D is the weakest on both indicators, with the highest IGD+ and the lowest HV (Table 6.2).

Table 6.2: Suction-cup results: Knob.

Algorithm	**IGD+ ↓**	**Archive HV ↑**	**Runtime (s)**
CMOPSO	0.0084 ± 0.0006	0.1449 ± 0.0018	56569.3 ± 2020.7
MOEA/D	0.0364 ± 0.0048	0.1168 ± 0.0158	82003.1 ± 3013.8
NSGA-II	0.0054 ± 0.0007	0.1617 ± 0.0075	63562.2 ± 587.6
NSGA-III	0.0055 ± 0.0023	$\mathbf{0.1639 \pm 0.0022}$	53231.9 ± 549.2
SMS-EMOA	$\mathbf{0.0052 \pm 0.0017}$	0.1602 ± 0.0072	46486.4 ± 1317.1

Connecting Rod

SMS-EMOA gives the lowest IGD+ and MOEA/D the highest, with NSGA-III second-highest. The archive HV does not separate the algorithms. All five algorithms overlap within their run-to-run spread, and the spreads of CMOPSO and MOEA/D are the widest. On this workpiece, the IGD+ ranking is therefore the only clear distinction between the algorithms (Table 6.3).

Table 6.3: Suction-cup results: Connecting rod.

Algorithm	IGD+ ↓	Archive HV ↑	Runtime (s)
CMOPSO	0.0162 ± 0.0020	0.2233 ± 0.0153	44538.9 ± 1172.3
MOEA/D	0.0307 ± 0.0038	0.2212 ± 0.0146	56479.5 ± 3251.1
NSGA-II	0.0172 ± 0.0011	$\mathbf{0.2243 \pm 0.0107}$	59928.5 ± 1524.4
NSGA-III	0.0224 ± 0.0016	0.2183 ± 0.0105	48552.1 ± 507.5
SMS-EMOA	$\mathbf{0.0096 \pm 0.0016}$	0.2210 ± 0.0093	40901.3 ± 458.9

Stanford Bunny

SMS-EMOA gives the lowest IGD+ and the highest archive HV on the Stanford Bunny, with NSGA-II a close second on both. CMOPSO and MOEA/D have the widest IGD+ spreads, so their positions are the least certain. NSGA-III lies between the leading pair and CMOPSO on both indicators, and MOEA/D is the weakest on both (Table 6.4).

Table 6.4: Suction-cup results: Stanford Bunny.

Algorithm	IGD+ ↓	Archive HV ↑	Runtime (s)
CMOPSO	0.0108 ± 0.0034	0.1068 ± 0.0021	81632.1 ± 439.4
MOEA/D	0.0189 ± 0.0035	0.1005 ± 0.0059	91043.0 ± 1149.3
NSGA-II	0.0049 ± 0.0010	0.1128 ± 0.0038	58394.7 ± 801.2
NSGA-III	0.0086 ± 0.0019	0.1114 ± 0.0025	113785.4 ± 1261.1
SMS-EMOA	$\mathbf{0.0029 \pm 0.0002}$	$\mathbf{0.1131 \pm 0.0039}$	52652.8 ± 776.8

6.1.4 Cross-Workpiece Summary

Figures 6.2, 6.3, and 6.4 show the three indicators for all algorithms and workpieces. Boxes show the median and interquartile range, red markers show the mean ± one sample standard deviation, and dots show individual runs ($n = 5$ per algorithm).

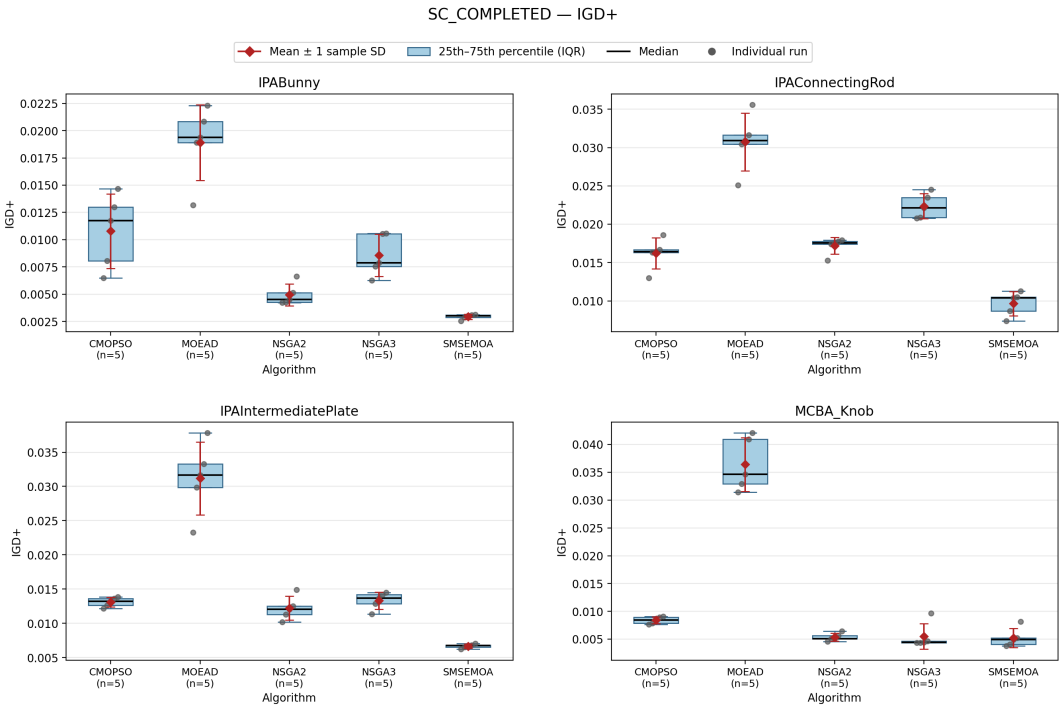


Figure 6.2: IGD+ per workpiece and algorithm ($n = 5$ runs per algorithm; lower is better). Values use the shared reference front of each workpiece.

Figure 6.2 shows that SMS-EMOA has the lowest mean IGD+ on all four workpieces, and MOEA/D the highest on all four. SMS-EMOA is also the most concentrated on the IPA Intermediate Plate and the Stanford Bunny, whereas on the Knob CMOPSO and NSGA-II have a smaller run-to-run spread, and on the Connecting rod NSGA-II does. NSGA-II is also tightly grouped on every workpiece. On the Knob, the mean IGD+ values of NSGA-II, NSGA-III, and SMS-EMOA lie within one standard deviation of each other, so these three are not clearly separated. The wide CMOPSO box on the Stanford Bunny and the wide MOEA/D boxes on several workpieces show that run-to-run variation is large for these two algorithms. The conclusion is that SMS-EMOA achieves the lowest mean IGD+ for the suction cup on all four workpieces, although its run-to-run spread is not the smallest on every workpiece, while MOEA/D is consistently the weakest.

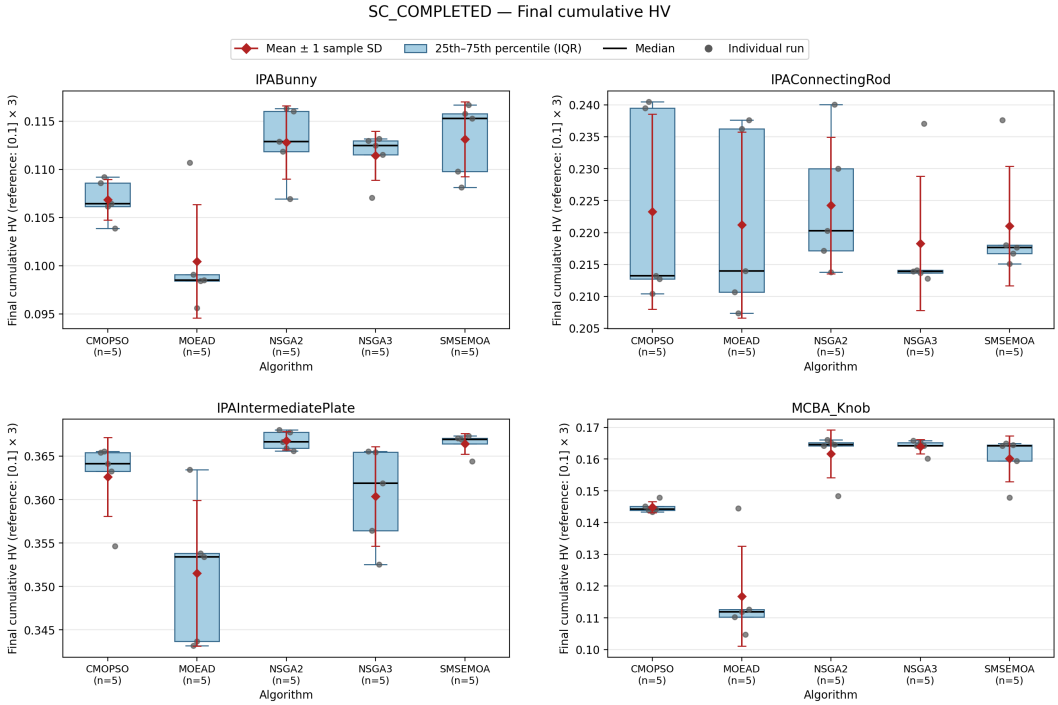


Figure 6.3: Final archive hypervolume per workpiece and algorithm ($n = 5$ runs per algorithm; higher is better).

Figure 6.3 shows that the boxes separate the algorithms on three of the four workpieces. On the Stanford Bunny, the boxes of NSGA-II, NSGA-III, and SMS-EMOA form a group that lies entirely above those of CMOPSO and MOEA/D. On the IPA Intermediate Plate, the boxes of NSGA-II and SMS-EMOA are high and tight, and CMOPSO lies just below them without overlap. NSGA-III has a wider box in the middle, and MOEA/D is lowest with the widest spread. On the Knob, the boxes of NSGA-II, NSGA-III, and SMS-EMOA are high and overlap, CMOPSO lies clearly below this group, and MOEA/D is lowest, with one run well above its box. On the Connecting rod, by contrast, the boxes of all five algorithms overlap across nearly the same range, and the means lie within about one standard deviation of each other, so no algorithm is separated. The conclusion is that HV distinguishes the algorithms on the Stanford Bunny, the IPA Intermediate Plate, and the Knob, where MOEA/D is lowest on each. On the Connecting rod, HV does not distinguish them conclusively.

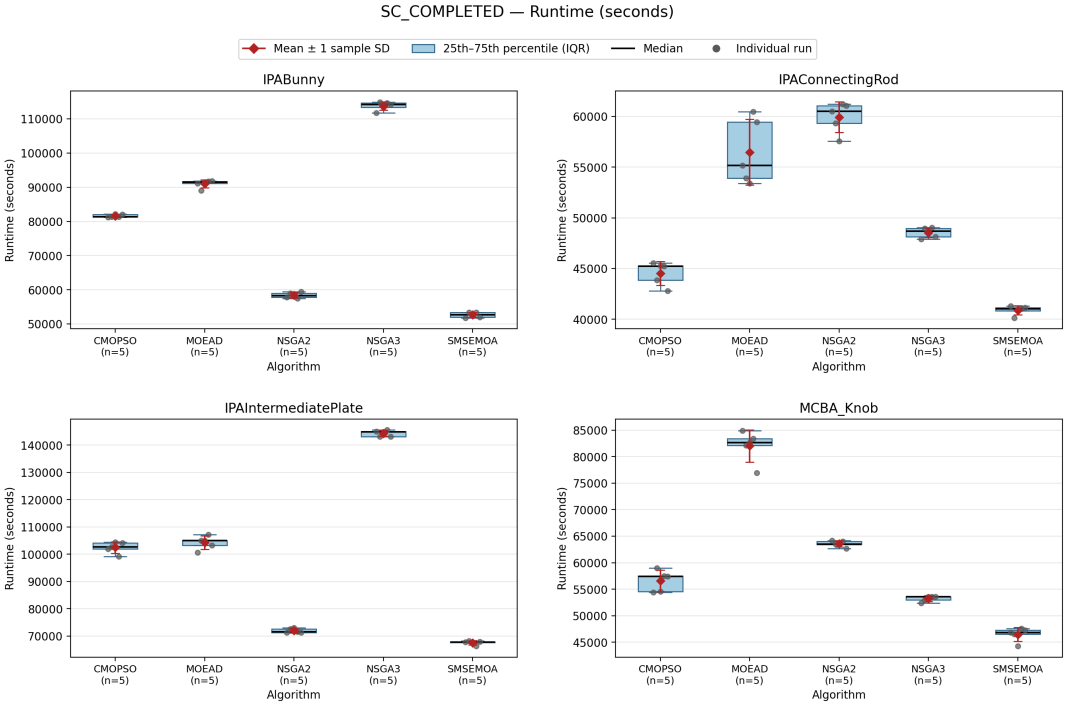


Figure 6.4: Wall-clock time per workpiece and algorithm ($n = 5$ runs per algorithm; seconds; indicative only, as the runs used different GPUs).

Figure 6.4 shows that run-to-run spread is small for most algorithm–workpiece combinations. The ordering of the algorithms changes between workpieces. For example, NSGA-III is the slowest on the Stanford Bunny and the IPA Intermediate Plate, while NSGA-II is the slowest on the Connecting rod and MOEA/D on the Knob. Because the runs used different GPUs, these differences cannot be separated from hardware effects. The figure is therefore used only to describe the computational cost, and no runtime ranking is drawn.

6.1.5 Hypervolume Progression

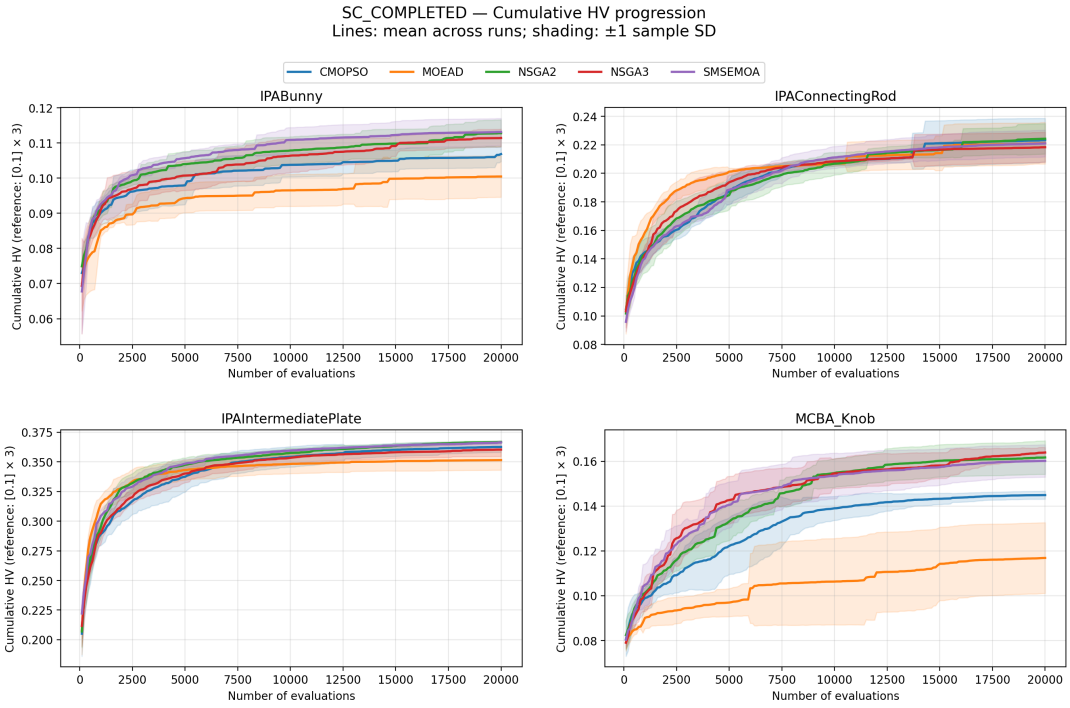


Figure 6.5: Cumulative archive hypervolume over evaluations, per workpiece and algorithm (mean across $n = 5$ runs; shading: ± 1 sample SD).

Figure 6.5 shows that most curves rise steeply in the first part of the run and then flatten gradually. Several curves are still rising at the end of the budget, particularly on the Stanford Bunny and the IPA Intermediate Plate. The final HV values therefore reflect the 20,000-evaluation budget as well as the algorithm quality.

On the Stanford Bunny, MOEA/D is briefly ahead at the start, but it falls behind early and then stays at the lowest level, with little change after the middle of the run. SMS-EMOA and NSGA-II lead for most of the run, with NSGA-III and CMOPSO following. On the Connecting rod, MOEA/D leads during the early phase, but the other algorithms overtake it, and all curves merge into a narrow band from about the middle of the run. CMOPSO shows a late step in the final part of the run. On the IPA Intermediate Plate, the curves stay close together throughout. MOEA/D is consistently lowest, and SMS-EMOA and NSGA-II are highest at the end while still rising. On the Knob, NSGA-II, NSGA-III, and SMS-EMOA rise together within a narrow band, and NSGA-III is still increasing at the end. CMOPSO

reaches a plateau earlier at a lower level. MOEA/D rises slowly, shows a step in the middle of the run, and has a very wide band, which indicates that its runs diverge.

The conclusion is that the final ranking of the algorithms is mostly determined in the second half of the run. The early lead of MOEA/D on the Stanford Bunny and the Connecting rod does not persist.

6.2 Two-Finger Gripper Results

6.2.1 Experimental Settings

Each evolutionary algorithm was run three times per workpiece, with a population of 100, 100 generations, and a budget of 10,000 evaluations. The problem has four objectives, one grasp-quality KPI per predefined grasp orientation. The archive HV and the reference point are defined as for the suction-cup gripper. The IGD+ reference front of each workpiece is pooled from the final solutions of all five algorithms and three runs (15 runs) on that workpiece.

6.2.2 Individual Results per Workpiece

IPA Intermediate Plate

NSGA-II has the highest archive HV, closely followed by SMS-EMOA. SMS-EMOA gives the lowest IGD+, with NSGA-II within the run-to-run spread. CMOPSO and MOEA/D lie lower in HV, and MOEA/D is the weakest on both indicators, with the highest IGD+. NSGA-III has the widest HV spread on this workpiece (Table 6.5).

Table 6.5: Two-finger results: IPA Intermediate Plate.

Algorithm	**IGD+ ↓**	**Archive HV ↑**	**Runtime (s)**
CMOPSO	0.0310 ± 0.0094	0.0915 ± 0.0038	25014.9 ± 7.1
MOEA/D	0.0438 ± 0.0155	0.0851 ± 0.0061	66839.0 ± 20.3
NSGA-II	0.0148 ± 0.0020	$\mathbf{0.1138 \pm 0.0010}$	41697.5 ± 12602.0
NSGA-III	0.0233 ± 0.0077	0.1082 ± 0.0120	37865.6 ± 25243.9
SMS-EMOA	$\mathbf{0.0140 \pm 0.0044}$	0.1110 ± 0.0071	67543.7 ± 11.6

Knob

NSGA-II has the highest archive HV and the lowest IGD+, with CMOPSO close on IGD+. SMS-EMOA is second on HV but sits in the middle on IGD+. NSGA-III has the highest IGD+. MOEA/D has the lowest HV, clearly separated from the other four algorithms, which lie close together (Table 6.6).

Table 6.6: Two-finger results: Knob.

Algorithm	IGD+ ↓	Archive HV ↑	Runtime (s)
CMOPSO	0.0050 ± 0.0014	1.3491 ± 0.0072	27087.4 ± 1.2
MOEA/D	0.0075 ± 0.0004	1.3268 ± 0.0057	24614.5 ± 211.5
NSGA-II	$\mathbf{0.0049 \pm 0.0005}$	$\mathbf{1.3572 \pm 0.0083}$	46510.5 ± 13589.5
NSGA-III	0.0084 ± 0.0014	1.3461 ± 0.0065	38328.5 ± 24973.1
SMS-EMOA	0.0062 ± 0.0016	1.3550 ± 0.0098	75822.1 ± 13.0

Connecting Rod

NSGA-II is best on both indicators, with CMOPSO close on IGD+ and SMS-EMOA close on HV. The top three algorithms in HV differ little. NSGA-III has the second-highest IGD+. MOEA/D is clearly the weakest on both indicators (Table 6.7).

Table 6.7: Two-finger results: Connecting rod.

Algorithm	IGD+ ↓	Archive HV ↑	Runtime (s)
CMOPSO	0.0141 ± 0.0007	0.0309 ± 0.0003	27849.4 ± 1.7
MOEA/D	0.0281 ± 0.0033	0.0259 ± 0.0003	72605.3 ± 14.3
NSGA-II	$\mathbf{0.0138 \pm 0.0009}$	$\mathbf{0.0323 \pm 0.0003}$	46526.8 ± 14061.9
NSGA-III	0.0198 ± 0.0023	0.0318 ± 0.0008	38407.6 ± 24954.0
SMS-EMOA	0.0151 ± 0.0020	0.0320 ± 0.0003	30505.6 ± 7.9

Stanford Bunny

SMS-EMOA gives the lowest IGD+ and NSGA-II the highest archive HV. Neither algorithm dominates the other across both indicators, since SMS-EMOA lies within the HV spread of NSGA-II. CMOPSO and NSGA-III lie in the middle, with the IGD+ of NSGA-III clearly higher. MOEA/D is the weakest on both indicators, with a much lower HV than all other algorithms (Table 6.8).

Table 6.8: Two-finger results: Stanford Bunny.

Algorithm	IGD+ ↓	Archive HV ↑	Runtime (s)
CMOPSO	0.0300 ± 0.0031	0.5937 ± 0.0009	25498.5 ± 3.0
MOEA/D	0.0529 ± 0.0053	0.5406 ± 0.0058	24111.3 ± 113.4
NSGA-II	0.0296 ± 0.0041	$\mathbf{0.5999 \pm 0.0078}$	43799.9 ± 13306.3
NSGA-III	0.0410 ± 0.0024	0.5955 ± 0.0041	38078.9 ± 25104.9
SMS-EMOA	$\mathbf{0.0248 \pm 0.0024}$	0.5997 ± 0.0067	27821.6 ± 0.7

6.2.3 Cross-Workpiece Summary

Figures 6.6, 6.7, and 6.8 show the three indicators for all algorithms and workpieces. Boxes show the median and interquartile range, red markers show the mean ± one sample standard deviation, and dots show individual runs ($n = 3$ per algorithm).

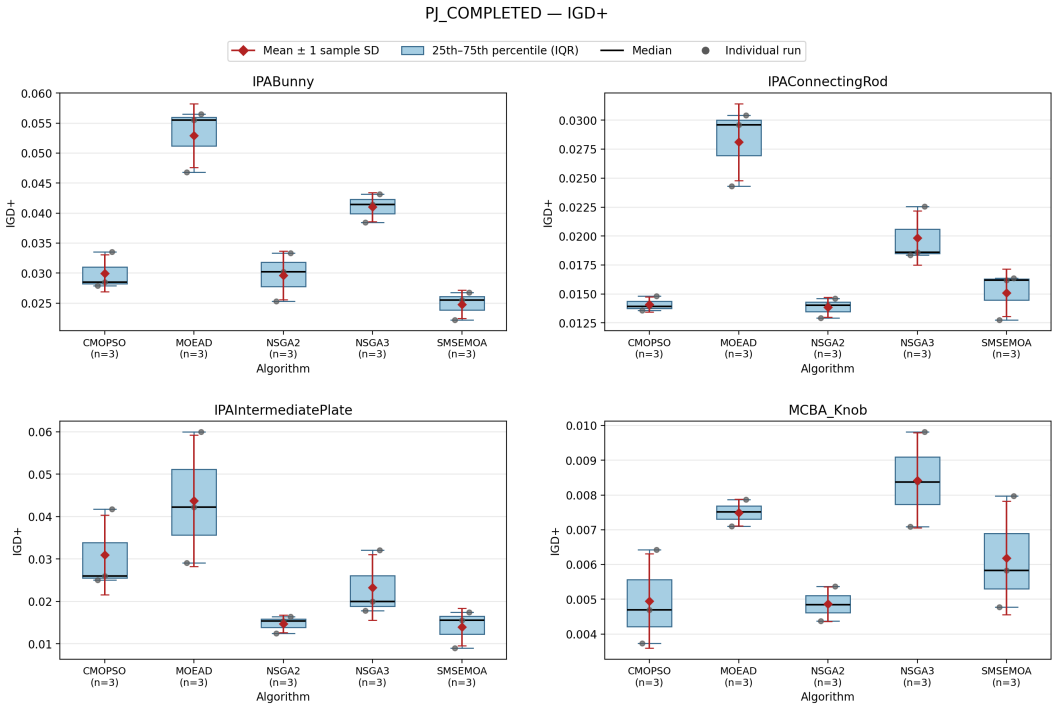


Figure 6.6: IGD+ per workpiece and algorithm ($n = 3$ runs per algorithm; lower is better). Values use the shared reference front of each workpiece.

Figure 6.6 shows that MOEA/D has the highest IGD+ on the Stanford Bunny, the Connecting rod, and the IPA Intermediate Plate, with its box clearly above the others on the first

two. On the Knob, NSGA-III has the highest IGD+, and MOEA/D lies above the lower group formed by CMOPSO and NSGA-II, whose boxes overlap. On the Connecting rod, CMOPSO and NSGA-II form the lowest group, with their boxes overlapping. On the IPA Intermediate Plate, the NSGA-II and SMS-EMOA boxes form the lowest group, while the CMOPSO and MOEA/D boxes are wide and only partly separated from them. SMS-EMOA is lowest on the Stanford Bunny. The conclusion is that no algorithm is lowest on all four workpieces. MOEA/D is the highest on three of the four and lies in the upper group on the fourth, while NSGA-II is in the lowest group on three of the four.

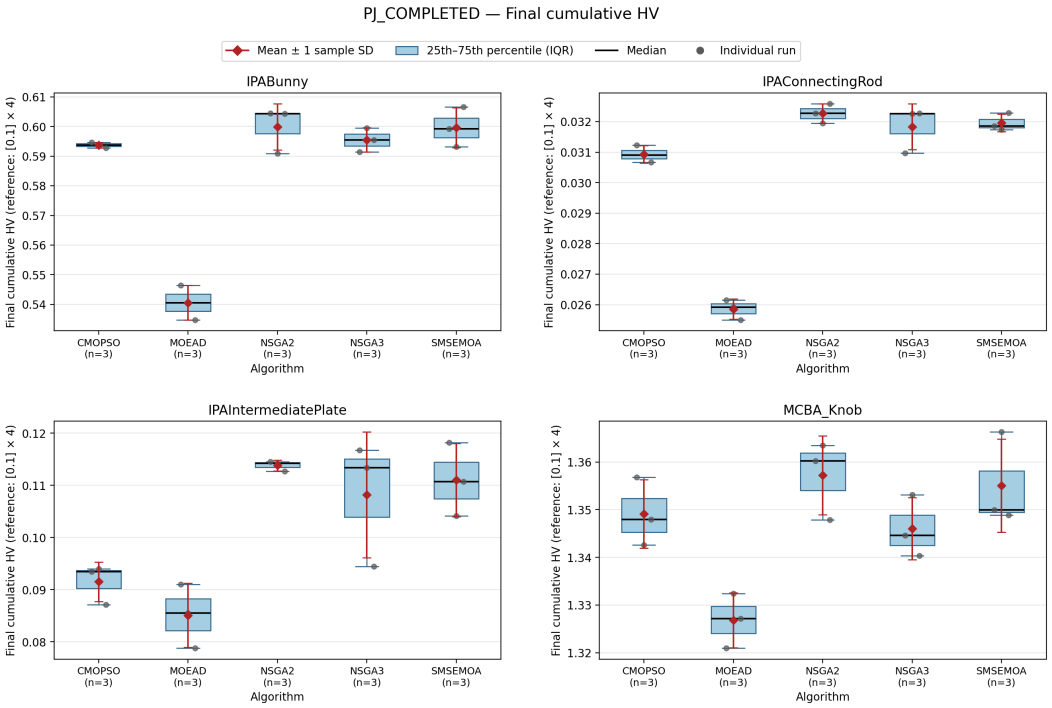


Figure 6.7: Final archive hypervolume per workpiece and algorithm ($n = 3$ runs per algorithm; higher is better).

Figure 6.7 shows that NSGA-II is highest or tied for highest on every workpiece, and MOEA/D is lowest on every workpiece, separated from the other algorithms on the Bunny and the Connecting rod. On the Connecting rod, NSGA-II, SMS-EMOA, and NSGA-III are close and tight. On the IPA Intermediate Plate, NSGA-II is tight and highest, while NSGA-III and SMS-EMOA have wide boxes that overlap with the middle group. On the Knob, SMS-EMOA has a wide box close to NSGA-II. The conclusion is that NSGA-II achieved the highest mean HV within each workpiece, and MOEA/D the least.

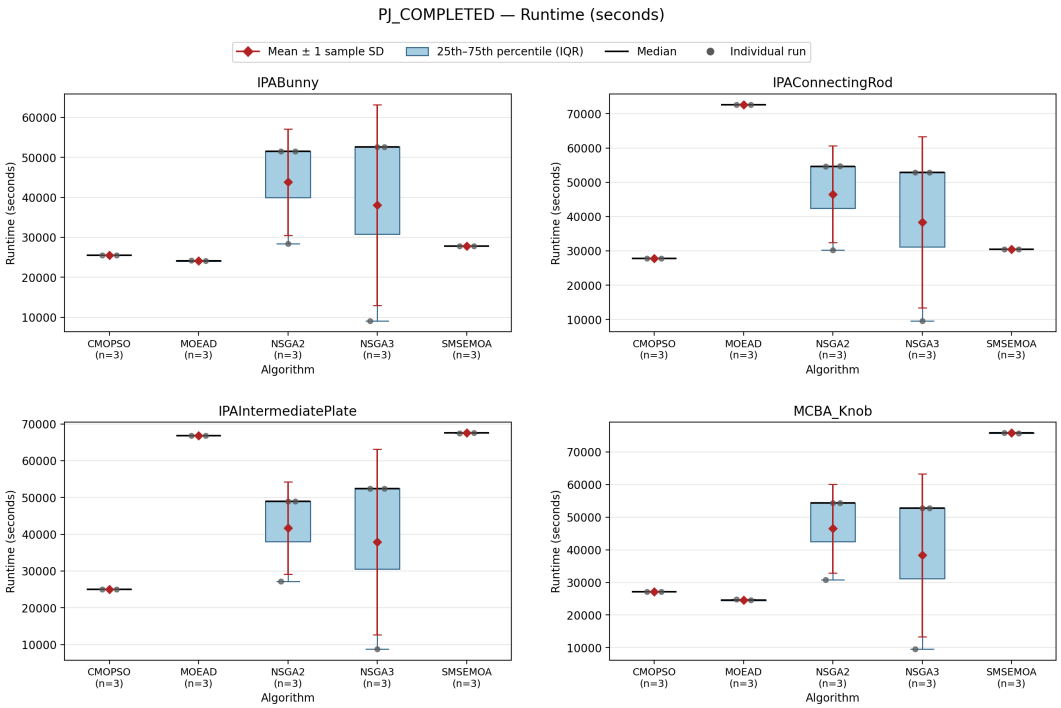


Figure 6.8: Wall-clock time per workpiece and algorithm ($n = 3$ runs per algorithm; seconds; indicative only, as the runs used different GPUs).

Figure 6.8 shows that CMOPSO and SMS-EMOA have narrow runtime spreads. NSGA-II and NSGA-III show a split pattern on each workpiece. Most of their runs are long, while one run is much shorter. The short runs are most likely caused by hardware differences, as stated above, but this cannot be verified from the data. MOEA/D is among the slowest on the Connecting rod and the IPA Intermediate Plate, and among the fastest on the Stanford Bunny and the Knob. The conclusion is that the runtime pattern is not interpretable as an algorithm property, and no runtime ranking is drawn.

6.2.4 Hypervolume Progression

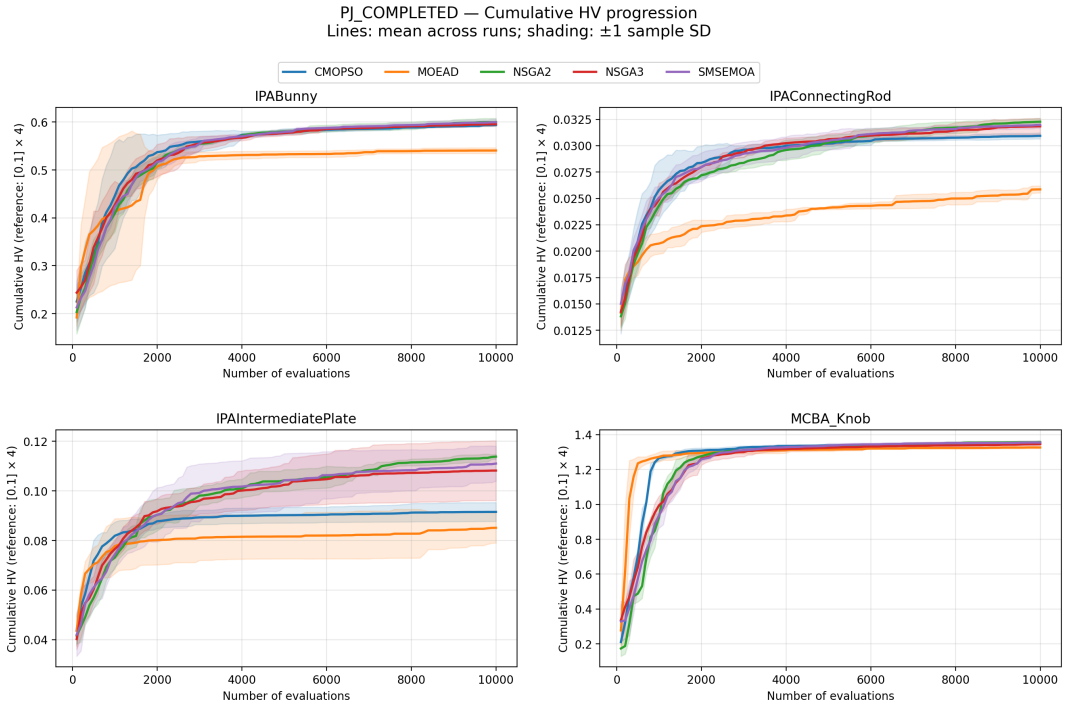


Figure 6.9: Cumulative archive hypervolume over evaluations, per workpiece and algorithm (mean across $n = 3$ runs; shading: ± 1 sample SD).

Figure 6.9 shows that the Knob converges fastest. All algorithms reach a close, flat final level after the early part of the run, with narrow bands. MOEA/D rises fastest at the start on the Knob, but the other algorithms catch up and MOEA/D ends lowest, so its early advantage is not kept.

On the Stanford Bunny, MOEA/D rises quickly in the early phase, but it flattens at the lowest level at around the first fifth of the run. The other four algorithms overlap and keep rising slowly until the end, converging to a narrow band from the middle of the run. On the Connecting rod, MOEA/D grows slowly and stays far below the others. The other four rise steeply early and keep improving until the end. NSGA-II ends highest and is still rising. On the IPA Intermediate Plate, the trajectories differ most. CMOPSO and MOEA/D flatten early, with CMOPSO stagnating from the middle of the run. NSGA-II keeps rising until the end and ends highest. NSGA-III and SMS-EMOA are also still rising at the end, with wide bands.

The conclusion is that for the Knob, the hypervolume curves plateaued within the available budget, the Connecting rod and the Stanford Bunny are close to convergence, and the IPA Intermediate Plate is not converged for NSGA-II, NSGA-III, and SMS-EMOA at 10,000 evaluations. The early speed of MOEA/D does not translate into the final HV on any workpiece.

6.3 Qualitative Inspection of Selected Solutions

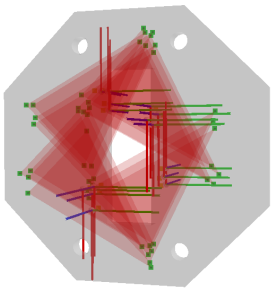
This section complements the quantitative results with a visual inspection of selected Pareto solutions. One solution is shown for each gripper–workpiece combination. For each instance, a run among the best or second-best runs across the compared algorithms was selected using either final archive hypervolume or IGD+. A solution from that run was then selected manually to favor high values across the objectives. This selection represents a subjective compromise rather than the result of a formal decision-making criterion.

The examples illustrate individual trade-offs and are not necessarily representative of the complete Pareto fronts. Since the images are generated using the same evaluation pipelines as the optimization, they do not independently validate those pipelines or establish physical grasp feasibility.

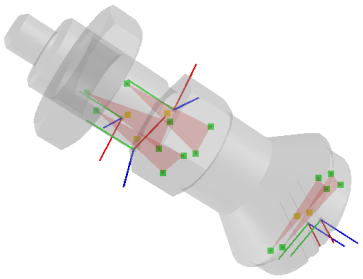
The objective values below are reported in the natural maximization convention. Values stored by the evolutionary framework were negated for minimization and have therefore been negated again for presentation. Decision variables and objective values are rounded.

6.3.1 Qualitative Inspection of Multi-Suction Cup Gripper

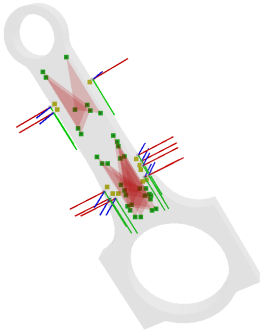
Figure 6.10 shows the retained grasp sets for four selected suction-cup layouts. Green square markers indicate the suction-cup positions, yellow markers indicate the tool center point (TCP), and the red, green, and blue lines indicate the axes of the grasp pose. The translucent triangles connect the three cup positions at each grasp pose. The figures show the grasp sets used to calculate diversity and normalized grasp count explained in Section 5.2.3.



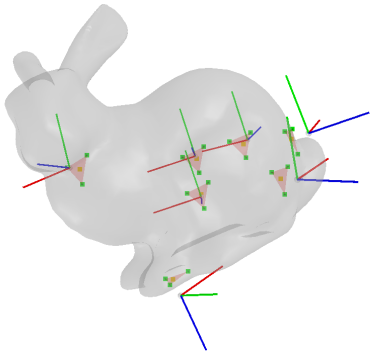
(a) IPA Intermediate Plate; SMS-EMOA



(b) Knob; NSGA-III



(c) Connecting rod; SMS-EMOA



(d) Stanford Bunny; SMS-EMOA

Figure 6.10: Retained grasp sets for selected multi-suction-cup solutions. Each translucent triangle connects the three cup positions at one grasp pose.

For each layout, the decision vector is $\mathbf{q}_{\mathrm{SC}} = (r,\alpha_1,\alpha_2,\alpha_3)$, where r is in meters and the angles are in radians. The objective vector is $(D,\hat{n}_{\mathrm{g}},\hat{A})$:

- **IPA Intermediate Plate:** $\mathbf{q}_{\mathrm{SC}} \approx (0.023124, 3.2962, 0.7594, 4.5371)$; objectives $(0.4193, 1.0000, 0.1965)$; 15 retained grasps.
- **Knob:** $\mathbf{q}_{\mathrm{SC}} \approx (0.018171, 1.5032, 4.3042, 4.9955)$; objectives $(0.5252, 0.3333, 0.0757)$; 5 retained grasps.
- **Connecting rod:** $\mathbf{q}_{\mathrm{SC}} \approx (0.015658, 2.6149, 1.7993, 4.4834)$; objectives $(0.4704, 0.7333, 0.0531)$; 11 retained grasps.
- **Stanford Bunny:** $\mathbf{q}_{\mathrm{SC}} \approx (0.005529, 2.4223, 0.4289, 4.8284)$; objectives $(0.4419, 0.4667, 0.0135)$; 7 retained grasps.

The grasp counts correspond to the normalized grasp-count objective with $K_{\text{clst}} = 15$. The figures show grasps in different regions of each workpiece. Some grasps are concentrated in particular areas, as visible on the connecting rod, while the plate and Bunny examples show grasps across multiple surface regions.

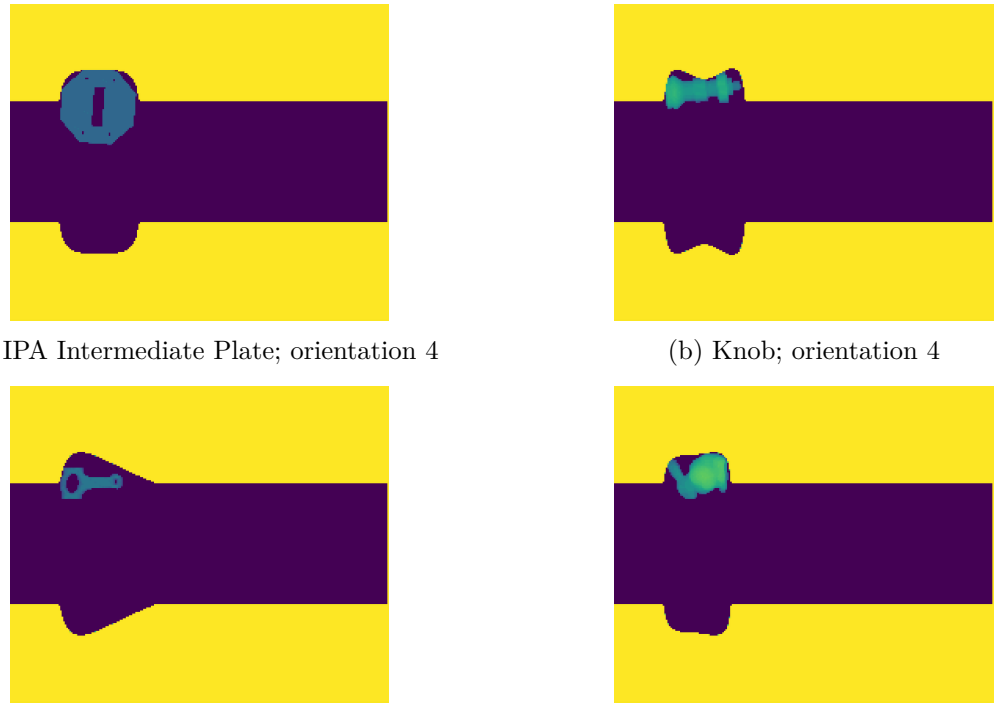
The positions and pose axes provide an intuitive impression of spatial and orientational diversity. However, the diversity objective combines normalized pairwise measures, so its numerical value cannot be determined from the images alone. Similarly, the triangles illustrate the cup arrangement, but their apparent size depends on the viewing projection.

The figures illustrate the grasp sets produced by the computational evaluation. They do not establish that the layouts will form seals or withstand external loads in a physical grasp.

6.3.2 Qualitative Inspection of Two-Finger Gripper

Figure 6.11 shows combined occupancy projections for the selected parallel-jaw solutions. For each workpiece, the displayed image corresponds to the predefined orientation with the highest proximity KPI for that solution.

As described in Section 4.2.3, the workpiece voxel grid is shifted using the selected left-jaw index and combined with the gripper grid in its modeled pose. The image is obtained by summing occupied voxels along the gripper length axis and its colors therefore represent projected occupancy rather than proximity scores.



(a) IPA Intermediate Plate; orientation 4

(b) Knob; orientation 4

(c) Connecting rod; orientation 1

(d) Stanford Bunny; orientation 4

Figure 6.11: Combined occupancy projections for selected NSGA-II parallel-jaw solutions. Each image shows the orientation with the highest proximity KPI for the corresponding solution.

The decision vector is $\mathbf{q}_{\mathrm{PJ}} = (\ell_{\mathrm{gn}}, h_1, h_2, h_3)$, with all four parameters in millimeters. The objective vector contains the proximity KPIs (k_1, k_2, k_3, k_4) in the configured orientation order:

- **IPA Intermediate Plate:** $\mathbf{q}_{\mathrm{PJ}} \approx (26.4661, 9.7092, 9.9724, 9.7111)$; objectives $(0.4605, 0.2966, 0.3529, 0.5735)$.
- **Knob:** $\mathbf{q}_{\mathrm{PJ}} \approx (27.2187, 9.6728, 7.1746, 9.9716)$; objectives $(0.9413, 0.8044, 0.7961, 1.0045)$.
- **Connecting rod:** $\mathbf{q}_{\mathrm{PJ}} \approx (31.7062, 9.9189, 7.1617, 3.4176)$; objectives $(0.4033, 0.1058, 0.1187, 0.1002)$.
- **Stanford Bunny:** $\mathbf{q}_{\mathrm{PJ}} \approx (22.3557, 8.7552, 9.2349, 9.9188)$; objectives $(0.5210, 0.4857, 0.7231, 0.7739)$.

The objective vectors show that a contour does not necessarily obtain the same proximity score in every orientation. For example, the selected connecting-rod contour has a higher

score in orientation 1 than in its other three orientations. The scores are scaled geometric KPIs and are interpreted within each workpiece problem.

The projections provide a qualitative view of the voxelized arrangements. They do not distinguish the workpiece from the gripper in the combined occupancy and do not show both jaws moved to their selected positions. Each image also shows only the highest-scoring orientation of the selected solution. The images therefore do not independently establish contact, collision-free placement, or grasp stability.

6.4 Bayesian Optimization Results

Bayesian optimization was carried out in two attempts. The main attempt was terminated due to its runtime and did not produce any result files. Subsequently, a minimal test was performed on a single gripper–workpiece pair to obtain indicative results. This test consists of a single run ($n = 1$) and serves as a starting point for future work rather than as a basis for a conclusive evaluation. Its limitations are discussed in Section 7.2.

6.4.1 Main Attempt

The runs were configured for 100 batches of 10 trials, corresponding to a planned budget of 1,000 evaluations. Four workpiece optimizations ran simultaneously in separate processes. The runs were stopped manually after more than 90 h; the reported progress was approximately 450 trials per run. No final result files were produced.

The logs indicate that candidate generation was time-consuming. Batches of ten trials were marked completed within seconds after their generation, whereas intervals containing Ax and BoTorch warnings lasted substantially longer. This suggests that model fitting and acquisition optimization contributed to the runtime, although the logs do not provide separate timings for these steps.

Several factors may have contributed to this cost. For four decision variables and a batch of ten candidates, joint acquisition optimization would involve $10 \times 4 = 40$ candidate coordinates. In addition, as observations accumulate, fitting a Gaussian process becomes more expensive (Section 2.3.7). The four processes maintained separate models and search spaces, but competition for shared computational resources may also have increased runtime.

The logs further report numerical and optimization warnings:

- Repeated `A not p.d., added jitter of 1.0e-08` warnings show that diagonal jitter was added during matrix factorization. Such warnings can arise from numerical ill-conditioning, although their cause cannot be established from the logs alone.
- A `gen_candidates_scipy` optimization reported status `ABNORMAL` and was retried with new initial conditions. Such retries can add to candidate-generation time.
- Ax reported a `MultiObjective` without objective thresholds and consequently winsorized each objective separately. The effect of this configuration on acquisition optimization and runtime was not measured.

The runs were therefore interrupted because of the time required, rather than an unrecoverable exception. Since the logs from concurrent processes are interleaved and do not isolate model-fitting, acquisition, and evaluation times, the relative contribution of each factor remains unresolved.

6.4.2 Minimal Test on the Connecting Rod

After the main attempt was stopped, a minimal test was run for the two-finger gripper on the Connecting rod only. It used a batch size of 5 and 100 batches, i.e. up to 500 evaluations, and was run once ($n = 1$). A total of 496 evaluations were completed. The problem, reference point, and archive-HV definition are the same as for the evolutionary algorithms in Section 6.2. Only the archive HV and the runtime are reported. No standard deviation is available because $n = 1$.

Table 6.9 adds the Bayesian run to the two-finger results on the Connecting rod. After 496 evaluations, the Bayesian run reaches an archive HV of 0.0281. This is about 87 % of the highest final HV (NSGA-II, 0.0323) and about 91 % of that of CMOPSO (0.0309). It is about 9 % higher than the final HV of MOEA/D (0.0259). The evolutionary algorithms used 10,000 evaluations, about 20 times more.

Table 6.9: Two-finger results on the Connecting rod, including the Bayesian minimal test. Evolutionary algorithms: $n = 3$ runs, 10,000 evaluations. Bayesian: $n = 1$ run, 496 evaluations, no standard deviation available for Bayesian.

Algorithm	**Archive HV ↑**	**Runtime (s)**	**Time per eval. (s)**
CMOPSO	0.0309 ± 0.0003	27849.4 ± 1.7	2.8
MOEA/D	0.0259 ± 0.0003	72605.3 ± 14.3	7.3
NSGA-II	$\mathbf{0.0323 \pm 0.0003}$	46526.8 ± 14061.9	4.7
NSGA-III	0.0318 ± 0.0008	38407.6 ± 24954.0	3.8
SMS-EMOA	0.0320 ± 0.0003	30505.6 ± 7.9	3.1
Bayesian ($n = 1$)	0.0281	66684.3	134.4

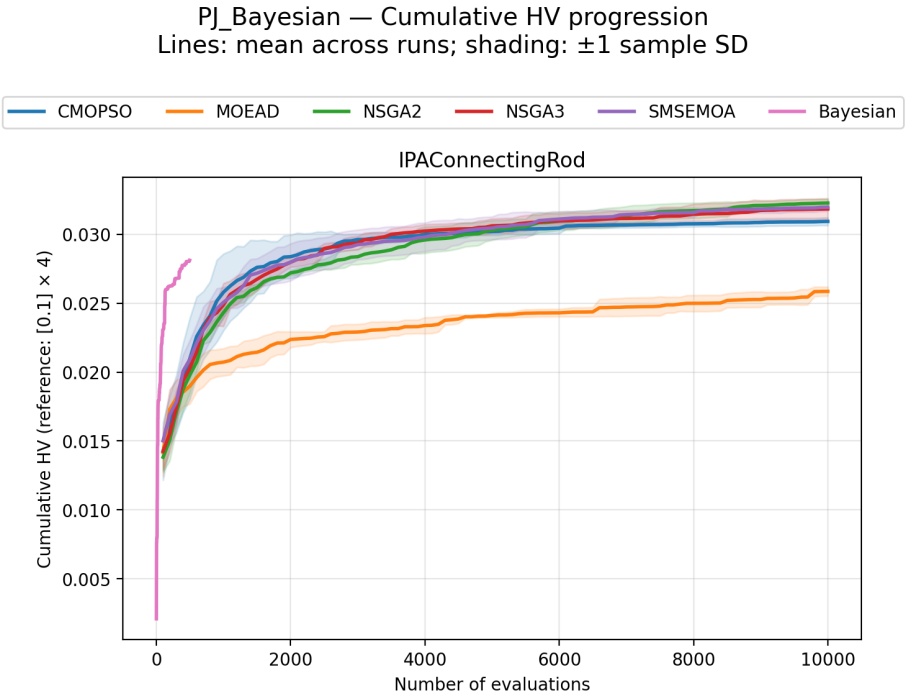


Figure 6.12: Cumulative archive hypervolume over evaluations on the Connecting rod (two-finger gripper). Evolutionary algorithms: mean across $n = 3$ runs, shading ± 1 sample SD. Bayesian: single run of 496 evaluations, no shading.

Figure 6.12 shows that the Bayesian curve rises much faster per evaluation than those of all evolutionary algorithms. It reaches an HV of about 0.026 within the first 100–150 evaluations, where the evolutionary algorithms are at about 0.014–0.017. The curve then increases slowly and ends at about 0.028 after 496 evaluations. At equal numbers of evaluations, the

Bayesian run is therefore above all evolutionary algorithms. CMOPSO, NSGA-II, NSGA-III, and SMS-EMOA exceed this value only after about 1,800 to 3,000 evaluations, and MOEA/D does not reach it within 10,000 evaluations.

The runtime of the test was 66,684 s (about 18.5 h), or about 134 s per evaluation. The evolutionary algorithms needed about 2.8 to 7.3 s per evaluation, so the Bayesian run is roughly 18 to 48 times more expensive per evaluation. The runs were carried out on different GPUs, so the measured times are not directly comparable. The runtimes are therefore indicative only and are not a conclusive result.

7 Summary

7.1 Conclusion

This thesis investigated multi-objective optimization methods for two gripper-design problems: the placement of three suction cups and the design of rigid parallel-jaw contours. Both were formulated as black-box problems and implemented through a common interface that separates candidate generation from problem-specific geometric evaluation.

NSGA-II, NSGA-III, MOEA/D, SMS-EMOA, and CMOPSO were evaluated on four workpieces, with each gripper–workpiece combination treated as a separate problem instance. Solution quality was assessed using IGD+ and cumulative archive hypervolume (HV), and runtime was recorded as an indication of computational cost. For the suction-cup problem, SMS-EMOA achieved the lowest mean IGD+ on all four workpieces, and NSGA-II was also among the leading methods. HV did not identify a clear leader on every workpiece. For the parallel-jaw problem, NSGA-II achieved the highest mean HV on all four workpieces, while the lowest mean IGD+ depended on the workpiece. MOEA/D generally performed less well than the leading methods. SMS-EMOA and NSGA-II are therefore promising choices for the investigated formulations and budgets, but no universally best algorithm was established.

The Ax/BoTorch Bayesian runs (batch size 10, four workpieces in parallel) were stopped after more than 90 hours, at approximately 450 trials per run, and produced no final results. A subsequent minimal test on the parallel-jaw gripper and the connecting rod (batch size 5, 496 evaluations, one run) reached an HV of 0.0281. This is about 87 % of the best evolutionary result (NSGA-II, 0.0323) with about 5 % of the evaluations. However, model fitting and candidate generation were very time-consuming, and repeating the optimization for several runs, further workpieces, and the suction-cup gripper would multiply this cost. Bayesian optimization therefore did not produce usable results in this work, whereas the population-based methods completed all reported experiments.

7.2 Limitations

The study used five independent runs per algorithm and workpiece for the suction-cup problem and three for the parallel-jaw problem, so the statistics are descriptive and small differences should be interpreted cautiously. Algorithm parameters were not tuned individually, and some HV curves were still rising at the end of the evaluation budget.

The true Pareto fronts are unknown. IGD+ was therefore computed against an empirical reference front built from the final solutions of the compared runs, and it measures performance relative to those runs. It also evaluates the final solution sets, whereas the reported HV uses a cumulative archive, so the two indicators do not assess exactly the same sets.

Runtimes are indicative only because the runs used GPUs with different computing capacities. The seed values and hardware information were not stored in the `results.json` metadata, so the GPU of each run cannot be identified and the runs cannot be reproduced exactly from the stored results. The Bayesian results are limited to a minimal single-run test ($n = 1$) on one gripper–workpiece pair, with a smaller batch size (5) than the stopped main attempt (10) and about 5 % of the evaluation budget of the evolutionary algorithms. They were compared only through HV, and their variability and their behavior at larger budgets or on other workpieces are unknown. In the main attempt, the four parallel processes shared computational resources, and the logs did not separate model fitting, candidate generation, and evaluation times. The causes of the long runtime therefore remain unresolved, and the outcome cannot be generalized to Bayesian optimization as a whole.

The suction-cup parameterization does not explicitly enforce a minimum distance between cup centers. Consequently, some candidate layouts may contain overlapping cups and may not be physically realizable. The radial bounds alone do not prevent overlap because the angular positions are optimized independently.

The evaluation is entirely computational. The suction-cup seal model and the voxel-based proximity and collision calculations of the parallel-jaw evaluator do not establish physical grasp performance, and the optimized configurations were not validated on real grippers. Manufacturing effects, material behavior, and contact forces were not assessed. Only four workpieces and restricted design spaces were investigated.

7.3 Future Work

The Bayesian comparison should be repeated under controlled hardware conditions with separate timing of model fitting, candidate generation, and geometric evaluation. The minimal test on the connecting rod can serve as a starting point and should be extended to several independent runs, the remaining workpieces, and the suction-cup gripper. Settings such as objective thresholds and batch size should be investigated. Running the processes sequentially and saving intermediate results would help locate bottlenecks and preserve partial runs. Comparisons should consider solution quality at common evaluation counts as well as wall-clock time.

Future experiments should store the metadata needed for reproducibility in the result files, including the random seed of every run along with the GPU and CPU model. This would allow runs to be repeated and runtimes to be compared across hardware.

Further algorithm studies could use more repetitions, statistical tests, and systematic parameter studies. Hybrid and surrogate-assisted methods could reduce the number of expensive evaluations. Since the Bayesian run reached a high HV after few evaluations, a short Bayesian phase could, for example, initialize an evolutionary algorithm.

A non-overlap condition should be added to the suction-cup formulation, requiring the distance between every pair of cup centers to be at least the cup diameter, and existing layouts should be checked for overlap. The evaluations could also be extended with additional design variables and force-based criteria, and the voxel-based parallel-jaw calculations should be compared with direct collision tests. Finally, physical experiments with selected suction-cup layouts and jaw contours in a bin-picking setup could show whether better computational objectives lead to reliable grasping.

A Appendix

A.1 Use of Generative AI

The table below documents the use of generative AI during the preparation of this thesis.

Table A.1: Documentation of generative AI use

Work step	**AI system**	**Location**	**Description of use**
Text drafting and refinement	Claude Haiku and Claude Sonnet (Anthropic)	Chapters 1–7	Self-written text refined and restructured; alternative wording and LaTeX text generated.
Equations and mathematical notation	Claude Haiku and Claude Sonnet (Anthropic)	Relevant sections of Chapters 1–7	Self-written equations and mathematical descriptions refined and formatted in LaTeX.
Abstract	Claude Opus 5 (Anthropic)	Abstract	Abstract generated from the supplied thesis content.
Abbreviation list	Claude Opus 5 (Anthropic)	List of Abbreviations	Abbreviations extracted from the thesis and formatted as a LaTeX list.
Table formulation	GPT6	Tables in Chapters 5 and 6	Tables formulated and presented in LaTeX using the supplied data.
Figure code	Claude Haiku and Claude Sonnet (Anthropic)	Figures 5.1, 5.2, and 5.3	LaTeX code generated for the software architecture diagram, candidate-evaluation sequence, and optimization workflow.

Continued on next page

Table A.1 continued

Work step	AI system	Location	Description of use
Citation support	Consensus	Chapters 1–7; Bibliography	Consensus used to support the preparation of citations and references.
Bibliography review	Claude Opus 5 (Anthropic)	Bibliography	BibTeX entry types reviewed and possible bibliographic corrections suggested.
Review and wording	Claude Opus 5 (Anthropic)	Chapters 1–7	Thesis reviewed for language, consistency, and selected technical issues; corrections and alternative wording suggested.

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